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Reconstruction Techniques for Elliptic Problems on Overlapping and Disjoint Domains

Part of the Master Research Thesis in Applied Mathematics

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Abstract

This work presents reconstruction techniques for elliptic problems on overlapping and disjoint computational domains, motivated by applications in the Variational Boussinesq Model (VBM). The proposed methodology is based on identifying the homogeneous component introduced by artificial domain splitting and removing it from the independently computed local solutions, thereby recovering an approximation of the solution that would have been obtained on the original, undecomposed domain. Two reconstruction frameworks are investigated: an overlap-based approach, which utilizes information from shared regions between neighbouring subdomains, and a flux-based approach, which relies on interface continuity and flux conditions for disjoint domains. Numerical experiments in one and two dimensions demonstrate that both methods accurately recover the dominant interface-induced errors, achieving improvements of several orders of magnitude without requiring additional global elliptic solves. The results show that homogeneous reconstruction provides an efficient and robust strategy for multidomain elliptic problems and offers strong potential for future large-scale wave simulation applications.

Contents

1	Introduction	3
2	Overlapping Domains	4
2.1	One-Dimensional Problem	5
2.1.1	Error Analysis	8
2.1.2	2-Domains Test Case	10
2.2	Two-Dimensional Problem	12
2.2.1	Error Analysis	16
2.2.2	2-Domains Test Case	18
2.2.3	4-Domains Test Case	22
2.3	Conclusion and Remarks	25
3	Disjoint Domains	26
3.1	One-Dimensional Problem	26
3.1.1	Error Analysis	29
3.1.2	2-Domains Test Case	30
3.2	Two-Dimensional Problem	32
3.2.1	Homogeneous Reconstruction Space	33
3.2.2	Interface Continuity Conditions	33
3.2.3	Galerkin Reconstruction	34
3.2.4	Modal Decoupling	34
3.2.5	Conditioning and Error Analysis	35
3.2.6	Corrected Solutions	35
3.2.7	2-Domains Test Case	35
3.3	Conclusion and Remarks	37
4	Conclusion and Remarks	37
A	MARIN VBM-XMF Computational Grid	40
B	Two-Dimensional Homogeneous Modes and Modal Basis Functions	41
B.1	Homogeneous Problem	41
B.2	Separation of Variables	41
B.3	Transverse Eigenfunctions	42
B.4	Longitudinal Behaviour	42
B.5	Modal Representation of the Homogeneous Correction	42
B.6	Directional Basis Functions	43
B.6.1	Western Modes	43
B.6.2	Eastern Modes	43
B.6.3	Southern Modes	43
B.6.4	Northern Modes	43
C	Boundary-Source Representation and Modal Projection	44

1 Introduction

The simulation of surface gravity waves propagating over an incompressible and inviscid fluid can be considerably simplified by assuming the flow to be irrotational. Under this assumption, the governing equations can be formulated in terms of a velocity potential, leading to so-called potential-flow models. Such models are widely employed in coastal, offshore, and maritime engineering applications, including the design of wave fields for laboratory experiments, the analysis of wave penetration into harbours, the investigation of long-wave generation during loading and off-loading operations of tankers, and the assessment of green-water events on ships. The present work considers a real-time wave simulator based on the Variational Boussinesq Model (VBM). The VBM is a Boussinesq-type model derived from a variational formulation of the water-wave problem and has been developed to accurately describe nonlinear and dispersive wave phenomena while remaining computationally efficient (Refs. [7, 6]). In general, Boussinesq-type models achieve a reduction in dimensionality by formulating the governing equations at the free surface while incorporating the effects of the interior flow through suitable approximations. The numerical implementation considered in this work follows the VBM framework developed by Lawrence et al. (Ref. [7]) and builds upon the variational Boussinesq formulations and numerical techniques described by Klopman (Ref. [6]). In particular, the computational aspects of the model and its application to real-time wave simulation are closely related to the work of Van 't Wout, who investigated improvements of the linear solver used within an interactive wave model environment (Ref. [16]). The objective of the present study is to further improve the computational efficiency of the wave simulator by developing and analyzing hyperbolic reconstruction techniques for overlapping and disjoint computational domains.

The main idea is to partition large computational wave domains into smaller subdomains and, at each time step, solve a small linear system to reconstruct the solution in the vicinity of the interfaces shared by neighboring subdomains. Several domain decomposition techniques already exist for problems of this nature, particularly for equations related to the Helmholtz operator (Refs. [15, 3]). The connection with the present work stems from the fact that many wave-propagation problems can, after suitable transformations or separation of variables, be formulated as Helmholtz or Sturm–Liouville problems. Consequently, the treatment of artificial interfaces and the transmission of information between neighboring subdomains are common challenges shared by these formulations. Among the most widely used domain decomposition techniques are Schwarz-type methods, which couple adjacent subdomains through transmission conditions imposed at the interfaces. In particular, optimized Schwarz methods employing Robin or impedance transmission conditions have been shown to significantly improve convergence rates for wave-propagation and Helmholtz problems (Refs. [9, 4, 5]). More recent developments include non-overlapping domain decomposition methods based on high-order transmission operators, approximate Dirichlet-to-Neumann maps, and cross-point treatments, all of which aim to improve convergence while minimizing communication between neighboring subdomains (Refs. [12, 10]). Recent advances have also explored the incorporation of machine-learning techniques into domain decomposition methods. Deep Domain Decomposition Methods and related learning-based approaches employ neural networks to approximate local solution operators, transmission conditions, interface mappings, or coarse-grid corrections directly from data (Refs. [8, 17]). These methods seek to reduce communication costs and improve scalability by exploiting learned representations of the underlying solution structure. Although the methodology developed in the present work remains entirely physics-based and does not rely on data-driven training, these developments highlight the growing interest in accurately recovering interface information while reducing computational complexity.

The present work pursues a similar objective, but does so through analytical reconstruction techniques derived from the structure of the governing differential operator. Rather than relying on iterative exchanges of interface information, the proposed framework exploits the linearity of the governing equation and the associated superposition principle. For linear differential operators, any solution of a non-homogeneous problem may be expressed as the sum of a particular solution and a homogeneous solution (Ref. [13]). In particular, if $\mathcal{L}$ denotes the linear differential operator associated with the problem, then any solution u of the non-homogeneous equation can be decomposed as

$$u = u_p + u_h,$$

where u_p is a particular solution satisfying the non-homogeneous equation and u_h belongs to the kernel of $\mathcal{L}$, i.e.,

$$\mathcal{L}(u_h) = 0.$$

Consequently, the effects introduced by artificial domain partitioning may be interpreted as homogeneous corrections superimposed on the local subdomain solutions. This viewpoint forms the theoretical basis of the reconstruction methodology developed in this work. The reconstruction procedure is based on identifying the homogeneous component introduced by the artificial domain splitting and removing it from the local solutions, thereby recovering an approximation of the solution that would have been obtained on the original, undecomposed domain. The construction of these homogeneous corrections naturally leads to the analysis of the associated eigenvalue problem. Indeed,

homogeneous solutions are closely related to the eigenfunctions of the governing operator through the classical theory of Sturm–Liouville and elliptic boundary value problems (Ref. [13]). These eigenfunctions provide a natural basis for representing the interface error and the homogeneous correction. In one spatial dimension, the homogeneous solution is often characterized by a limited number of admissible modes. In two spatial dimensions, however, multiple eigenvalues and associated eigenfunctions may contribute simultaneously, resulting in a higher-dimensional reconstruction space. The theoretical framework therefore combines the decomposition into particular and homogeneous solutions with concepts from spectral theory and eigenfunction expansions (Ref. [13]). Two classes of domain decompositions are considered. For overlapping decompositions, the reconstruction relies exclusively on the independently computed subdomain solutions. Since neighboring subdomains share a common region, the overlap itself contains sufficient information to identify the homogeneous correction, eliminating the need for additional global solves or iterative interface procedures. For non-overlapping decompositions, the absence of a shared region requires supplementary interface information. In this case, the reconstruction is formulated in terms of the fluid flux across the artificial interfaces, from which the homogeneous correction can be inferred. The problem considered in this work is the elliptic boundary value problem

$$ku - \nabla \cdot (h \nabla u) = \nabla \cdot (p \nabla f), \quad (1)$$

posed on a spatial domain $\Omega \subset \mathbb{R}^d$, where $d \in \{1, 2\}$. Here, $u : \Omega \rightarrow \mathbb{R}$ denotes the unknown field, $h : \Omega \times [0, T] \rightarrow \mathbb{R}$ is a coefficient associated with the local water depth, $f : \Omega \times [0, T] \rightarrow \mathbb{R}$ is related to the free-surface elevation, and $p : \Omega \times [0, T] \rightarrow \mathbb{R}$ denotes a pressure-related quantity. The scalar parameter $k \in \mathbb{R}$ is assumed to be known.

In applications arising from the Variational Boussinesq Model, the quantities h , p , and f generally depend on time. Consequently, equation (1) must be solved repeatedly throughout the simulation, producing a sequence of elliptic problems with time-dependent coefficients and forcing terms. To assess the applicability of the proposed methodology, both overlapping and non-overlapping decompositions are investigated in one and two spatial dimensions. The analysis begins with two-domain configurations, which allow the reconstruction mechanism to be studied in a controlled setting, and is subsequently extended to multidomain arrangements. Special attention is devoted to the role of homogeneous corrections, the associated eigenvalue structure, and the propagation of interface information throughout the computational domain. The remainder of this report is organized as follows. Section 2 introduces the reconstruction framework for overlapping domains and investigates its performance for one- and two-dimensional test problems, including both two-domain and multidomain configurations. Section 3 considers the corresponding reconstruction problem for non-overlapping domains and examines the use of flux information at artificial interfaces. Finally, Section 4 summarizes the main findings and discusses directions for future theoretical and numerical developments.

2 Overlapping Domains

Domain decomposition techniques partition a computational domain into smaller subdomains that can be solved independently. In an overlapping decomposition, neighbouring subdomains share a common region in which information from multiple local solutions is simultaneously available. Unlike iterative domain decomposition methods, the present approach exploits this overlap region to directly reconstruct the error introduced by the artificial partitioning of the computational domain.

The notation adopted throughout this work follows the computational-grid convention used by the Variational Boussinesq Model (VBM) and the MARIN VBM-XMF implementation described by Klopman (Ref. [6]); see also Appendix A. In particular, the directions north, south, east, and west are used to identify neighbouring nodes, boundaries, and subdomains. These labels correspond exclusively to the grid orientation employed by the numerical implementation and should not be interpreted as geographical directions or as part of an Earth-fixed coordinate system.

The use of this notation is particularly natural within the VBM-XMF framework, where ghost nodes are introduced outside the physical domain to impose boundary conditions and improve the accuracy of the discretized elliptic operators. The reconstruction methodology developed in the present work follows a similar philosophy. Rather than using additional points to enforce physical boundary conditions, information already available near artificial interfaces is utilized to recover the solution that would have been obtained on the original undecomposed domain.

The governing equation is given by Eq. (1). Since the associated differential operator is linear, any solution may be decomposed into a particular solution and a homogeneous solution (Ref. [13]),

$$u = u^p + u^h,$$

where

$$ku^p - \nabla \cdot (h \nabla u^p) = \nabla \cdot (p \nabla f),$$

and

$$ku^h - \nabla \cdot (h \nabla u^h) = 0.$$

This decomposition forms the theoretical basis of the reconstruction procedure. Since the forcing term is unaffected by the artificial decomposition, the dominant effect of the imposed interface conditions is contained in the homogeneous component. The reconstruction procedure is based on identifying the homogeneous component introduced by the artificial domain splitting and removing it from the local solutions, thereby recovering an approximation of the solution that would have been obtained on the original, undecomposed domain.

2.1 One-Dimensional Problem

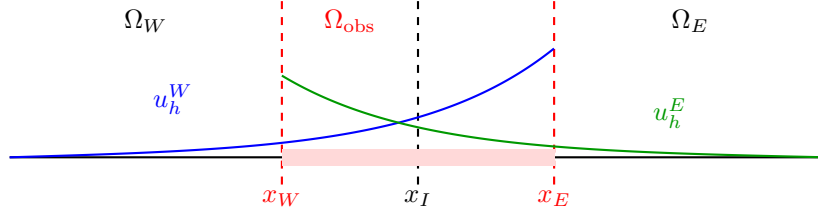


Figure 1: Overlapping-domain reconstruction. The subdomains Ω_W and Ω_E overlap in the observation domain $\Omega_{\text{obs}} = [x_W, x_E]$. The homogeneous corrections u_h^W and u_h^E satisfy artificial boundary conditions g_W and g_E at the overlap boundaries x_E and x_W , respectively. For constant coefficients, these corrections are represented by normalized hyperbolic-sine functions and decay away from the artificial boundaries. Since both subdomain solutions coexist throughout the overlap region, observations collected inside Ω_{obs} can be used to determine the unknown homogeneous amplitudes.

Consider the one-dimensional domain

$$\Omega = [x_0, x_N],$$

which is partitioned into a western subdomain Ω_W and an eastern subdomain Ω_E ,

$$\Omega_W = [x_0, x_E], \quad \Omega_E = [x_W, x_N].$$

When $x_W < x_E$, the decomposition is overlapping and the common interval

$$\Omega_{\text{obs}} = [x_W, x_E]$$

acts as an observation domain in which both independently computed subdomain solutions are simultaneously available. As illustrated in Figure 1, the overlap region contains information from both homogeneous corrections, allowing their amplitudes to be determined from local observations. For large overlaps, Ω_{obs} consists of the common portion shared by neighbouring subdomains. Within the VBM-XMF framework described in Appendix A, however, the overlap may be reduced to the ghost-node layer together with a limited number of neighbouring interior grid points. In the limiting case, it is theoretically possible to construct the observation domain using only a single layer of physical grid points from each subdomain, such that the outermost physical node of one subdomain is directly coupled to the ghost-node layer of the neighbouring subdomain.¹ Although such a minimal overlap is attractive from the perspective of memory usage and communication costs, it introduces two potential difficulties. First, the observation region may contain too few physical points to accurately characterize the homogeneous correction generated by the artificial interface. The reconstruction framework relies on identifying the spatial variation of the homogeneous corrections shown schematically in Figure 1. If only a very limited number of observations are available, the determination of the unknown correction amplitudes becomes increasingly sensitive to numerical errors. Second, the overlap region may contain too few points to adequately resolve the boundary-layer structure associated with the homogeneous correction. As illustrated in Figure 1, the influence of the artificial interface decays as the distance from the interface increases. Similar resolution requirements arise in singularly perturbed boundary-value problems, where accurate representation of rapidly varying boundary layers requires a sufficient number of grid points across the layer thickness (Ref. [14]). If only one or two physical grid points are available within the overlap region, the observed solution may not contain

¹When ghost nodes are employed, x_W and x_E may denote the western and eastern ghost-node locations. The observation domain then consists of the ghost-node layer together with adjacent physical nodes surrounding the artificial interface, allowing the reconstruction procedure to be incorporated directly into the MARIN VBM-XMF computational grid.

sufficient information to accurately recover the spatial structure of the homogeneous correction. A possible remedy consists of enriching the observation domain through interpolation. Additional virtual observation points may be introduced between neighbouring grid locations without requiring additional PDE solves. These interpolated observations provide a more refined description of the decay behaviour represented in Figure 1 and improve the robustness of the reconstruction procedure when only minimal overlap is available. Consequently, even configurations consisting of a ghost-node layer together with a single physical-node layer on either side of the interface may provide sufficient information for accurate reconstruction, provided that the variation of the homogeneous correction is adequately resolved.

After solving the governing equation independently on each subdomain, the local solutions can be decomposed into particular and homogeneous contributions,

$$\begin{aligned} u_W(x) &= u_p^W(x) + u_h^W(x), \\ u_E(x) &= u_p^E(x) + u_h^E(x). \end{aligned}$$

Since both subdomains solve the same differential equation with identical forcing terms, the corresponding particular solutions are expected to coincide within the overlap region,

$$u_p^W(x) = u_p^E(x), \quad \forall x \in \Omega_{\text{obs}}.$$

Consequently, the difference between the independently computed solutions is primarily associated with the homogeneous corrections,

$$u_W(x) - u_E(x) = u_h^W(x) - u_h^E(x), \quad x \in \Omega_{\text{obs}}. \quad (2)$$

The reconstruction problem therefore reduces to determining the homogeneous components generated by the artificial interface conditions.

For constant coefficients, the homogeneous equation

$$-(hu_x)_x + ku = 0 \quad (3)$$

reduces to

$$-hu_{xx} + ku = 0.$$

Seeking solutions of the form

$$u_h = e^{\lambda x}, \quad \lambda \in \mathbb{R},$$

yields the characteristic equation

$$h\lambda^2 - k = 0,$$

with roots

$$\lambda = \pm \sqrt{\frac{k}{h}}.$$

The general homogeneous solution may therefore be written as

$$u_h(x) = C_1 e^{\lambda x} + C_2 e^{-\lambda x}, \quad C_i \in \mathbb{R} \forall i$$

or equivalently as

$$u_h(x) = A \sinh(\lambda x) + B \cosh(\lambda x), \quad A, B \in \mathbb{R}.$$

The parameter λ characterizes the spatial decay of the homogeneous correction. Introducing

$$\delta = \frac{1}{\lambda} = \sqrt{\frac{h}{k}},$$

the quantity δ may be interpreted as the characteristic boundary-layer thickness associated with the artificial interface. Physically, it represents the distance over which the influence of the imposed interface condition propagates into the subdomain before becoming negligible. The homogeneous corrections illustrated in Figure 1 therefore behave as exponentially decaying boundary layers attached to the overlap boundaries.

Accurate reconstruction requires that the overlap region contains sufficient information to resolve this decay. Similar requirements arise in singularly perturbed boundary-value problems, where the numerical approximation of exponential boundary layers depends strongly on the relationship between the layer thickness and the grid spacing (Ref. [14]). If the overlap width is significantly smaller than δ , only a small portion of the decay profile is observed and the determination of the homogeneous amplitudes becomes increasingly sensitive to numerical errors.

Although the exponential representation spans the complete homogeneous solution space, the coefficients C_1 and C_2 do not directly reflect the boundary conditions imposed on the western and eastern subdomains. Consequently, they possess no immediate physical interpretation and must be determined together with additional boundary constraints. For the reconstruction procedure considered here, it is therefore advantageous to construct basis functions that automatically satisfy the physical boundary conditions of the original domain while remaining normalized at the artificial interfaces. This choice reduces the number of unknowns and allows the reconstruction coefficients to be interpreted directly as the amplitudes of the interface-induced homogeneous corrections.

Let x_0 and x_N denote the physical boundaries of the original domain. Define

$$\phi_W(x) = \frac{\sinh(\lambda(x - x_0))}{\sinh(\lambda(x_E - x_0))}, \quad (4)$$

and

$$\phi_E(x) = \frac{\sinh(\lambda(x_N - x))}{\sinh(\lambda(x_N - x_W))}, \quad (5)$$

which satisfy

$$\begin{aligned} \phi_W(x_0) &= 0, & \phi_W(x_E) &= 1, \\ \phi_E(x_N) &= 0, & \phi_E(x_W) &= 1. \end{aligned}$$

The homogeneous corrections are represented as

$$\begin{aligned} u_h^W(x) &= a_W \phi_W(x), \\ u_h^E(x) &= a_E \phi_E(x), \end{aligned}$$

where a_W and a_E denote the unknown interface amplitudes. Substituting these expressions into Eq. (2) gives

$$u_W(x_i) - u_E(x_i) = a_W \phi_W(x_i) - a_E \phi_E(x_i), \quad x_i \in \Omega_{\text{obs}}.$$

Evaluating Eq. (2) at two observation locations $x_1, x_2 \in \Omega_{\text{obs}}$ results in a system of two equations for the unknown amplitudes a_W and a_E . Introducing the observation vector

$$\mathbf{d} = \begin{bmatrix} u_W(x_1) - u_E(x_1) \\ u_W(x_2) - u_E(x_2) \end{bmatrix},$$

the amplitude vector

$$\mathbf{a} = \begin{bmatrix} a_W \\ a_E \end{bmatrix},$$

and the reconstruction matrix

$$\mathbf{H} = \begin{bmatrix} \phi_W(x_1) & -\phi_E(x_1) \\ \phi_W(x_2) & -\phi_E(x_2) \end{bmatrix}, \quad (6)$$

the reconstruction problem may be written compactly as

$$\mathbf{H}\mathbf{a} = \mathbf{d}. \quad (7)$$

Provided that $\mathbf{H}$ is nonsingular, the homogeneous amplitudes are obtained from

$$\mathbf{a} = \mathbf{H}^{-1}\mathbf{d}.$$

For the present one-dimensional formulation, invertibility of $\mathbf{H}$ is ensured when the observation points are distinct and the normalized basis functions ϕ_W and ϕ_E remain linearly independent within the overlap region. Once the amplitudes have been determined, the homogeneous corrections can be reconstructed throughout both subdomains,

$$\begin{aligned} u_h^W(x) &= a_W \phi_W(x), \\ u_h^E(x) &= a_E \phi_E(x), \end{aligned}$$

and subsequently removed from the local solutions,

$$\begin{aligned} \tilde{u}_W &= u_W - u_h^W, \\ \tilde{u}_E &= u_E - u_h^E. \end{aligned}$$

The previous derivation assumes that the particular solutions coincide exactly in the overlap region. In practice, however, finite-difference discretization, interpolation procedures, truncation errors, and round-off errors produce a small discrepancy,

$$u_p^W(x) = u_p^E(x) + \varepsilon(x),$$

where $\varepsilon(x)$ denotes the numerical mismatch. Consequently,

$$u_W(x) - u_E(x) = u_h^W(x) - u_h^E(x) + \varepsilon(x).$$

Even after removing the homogeneous corrections, a residual difference may remain between the reconstructed solutions. To reduce the influence of this discrepancy, a weighted blending procedure can be introduced, following ideas commonly employed in overlapping-domain and overset-grid methods (Refs. [15, 1]).

2.1.1 Error Analysis

The reconstruction framework developed above assumes that the mismatch between the independently computed subdomain solutions is entirely explained by the homogeneous corrections. For the continuous problem, this assumption follows from the uniqueness of the particular solution associated with the governing non-homogeneous boundary-value problem. In practice, however, numerical discretization introduces additional perturbations into the overlap observations.

Let τ_W and τ_E denote the discretization errors associated with the western and eastern subdomains, respectively. The computed solutions may then be expressed as

$$\begin{aligned} u_W &= u_p^W + u_h^W + \tau_W, \\ u_E &= u_p^E + u_h^E + \tau_E. \end{aligned}$$

Subtracting both expressions yields

$$u_W - u_E = (u_h^W - u_h^E) + (u_p^W - u_p^E) + (\tau_W - \tau_E).$$

Introducing the aggregate perturbation

$$\varepsilon = (u_p^W - u_p^E) + (\tau_W - \tau_E),$$

the overlap relation becomes

$$u_W - u_E = u_h^W - u_h^E + \varepsilon. \quad (8)$$

Equation (8) shows that the overlap mismatch is not explained solely by the homogeneous corrections but also contains a perturbation associated with numerical errors. Consequently, the amplitudes recovered from the reconstruction system differ from the exact amplitudes associated with the continuous problem.

Let

$$\mathbf{a}^* = \begin{bmatrix} a_W^* \\ a_E^* \end{bmatrix}$$

denote the exact amplitude vector, and let

$$\mathbf{a} = \begin{bmatrix} a_W \\ a_E \end{bmatrix}$$

denote the reconstructed amplitude vector obtained from Eq. (7). The amplitude error is then defined by

$$\Delta \mathbf{a} = \mathbf{a} - \mathbf{a}^*.$$

Using Eq. (7), the perturbation in the reconstructed amplitudes satisfies

$$\Delta \mathbf{a} = \mathbf{H}^{-1} \boldsymbol{\varepsilon},$$

where

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon(x_1) \\ \varepsilon(x_2) \end{bmatrix}.$$

The reconstruction accuracy therefore depends on two factors:

1. the magnitude of the perturbation $\boldsymbol{\varepsilon}$ present in the overlap observations;
2. the conditioning of the reconstruction matrix $\mathbf{H}$.

If $\mathbf{H}$ becomes nearly singular, even small perturbations in the overlap observations may lead to significant errors in the recovered amplitudes.

For a second-order finite-difference discretization, the local truncation error satisfies $\mathcal{O}(\Delta x^2)$. Assuming that both subdomains employ the same numerical scheme, the perturbation is expected to be bounded by $\mathcal{O}(\Delta x^2)$. Consequently,

$$\Delta \mathbf{a} \leq \mathcal{O}(\Delta x^2),$$

implying that the reconstruction convergence rate can at most be as fast as the order of accuracy of the underlying discretization.

A second source of error is associated with inadequate resolution of the homogeneous boundary layer. The characteristic boundary-layer thickness, δ determines the distance over which the influence of the artificial interface propagates into the computational domain. To accurately recover the decay behaviour illustrated in Figure 1, the overlap region must contain a sufficient number of observation points within this characteristic distance. Similar mesh-resolution requirements arise in singularly perturbed boundary-value problems involving exponential boundary layers (Ref. [14]). A useful measure is

$$N_\delta = \frac{\delta}{\Delta x} = \frac{\sqrt{h/k}}{\Delta x},$$

which represents the number of grid points contained within the characteristic decay region. Accurate reconstruction therefore requires

$$N_\delta \gg 1,$$

or equivalently,

$$\Delta x \ll \delta,$$

so that the exponential decay of the homogeneous correction is adequately resolved (Ref. [14]).

From the perspective of the reconstruction procedure, however, the principal requirement is not solely the number of points contained within the boundary layer. The selected observation points must also provide sufficient information to distinguish the homogeneous basis functions ϕ_W and ϕ_E . If the overlap width becomes small relative to the characteristic decay length, the basis functions become increasingly similar over Ω_{obs} , causing the reconstruction matrix $\mathbf{H}$ to become poorly conditioned. In this regime, even small numerical perturbations may generate large errors in the reconstructed amplitudes.

An a posteriori estimate of the reconstruction quality may be obtained from the residual

$$r(x) = u_W(x) - u_E(x) - (a_W\phi_W(x) - a_E\phi_E(x)).$$

This residual measures the portion of the overlap mismatch that is not explained by the reconstructed homogeneous correction. The quantities $\|r\|_2$, and $\|r\|_\infty$, therefore provide practical indicators of the combined discretization and reconstruction error. Small residual norms indicate that the discrepancy between the two subdomain solutions is well represented by the reconstructed homogeneous modes. Conversely, large residuals may indicate inadequate overlap width, insufficient boundary-layer resolution, inaccurate estimation of the decay parameter λ , rapidly varying coefficients, poor conditioning of the reconstruction matrix, or dominant discretization errors in the local subdomain solutions.

In such situations, a weighted averaging procedure can be used, and example is to reconstructed the solution as

$$u_R(x) = \omega(x)\tilde{u}_W(x) + (1 - \omega(x))\tilde{u}_E(x), \quad x \in \Omega_{\text{obs}},$$

where $\omega(x) \in [0, 1]$. A simple partition-of-unity weighting is

$$\omega(x) = \frac{x_E - x}{x_E - x_W},$$

which gradually shifts the contribution from the western solution to the eastern solution across the overlap region. This additional averaging step reduces the influence of local discretization errors and improves continuity throughout the reconstruction region. This provides an additional mechanism for reducing the influence of numerical discrepancies in the final reconstructed solution. Similar blending procedures are commonly employed in overlapping-domain and overset-grid methodologies to improve solution continuity across artificial interfaces (Refs. [15, 1]).

2.1.2 2-Domains Test Case

The objective is not to the present a proof-of-concept study on the previously introduced theory, the computational domain is defined as $\Omega = [0, 100]$. The governing equation is solved using with $k = 25$, together with the coefficient functions

$$\begin{aligned} h(x) &= 30 + 5 \cos(0.64x), \\ p(x) &= 10 + 64 \cos(0.64x), \\ f_x(x, t) &= 0.25 \cos(0.25x - 0.64t). \end{aligned}$$

The benchmark solution on the undecomposed domain is computed using 800 finite elements, while the western and eastern subdomains are discretized independently using 400 finite elements. The overlap region is reduced to the smallest possible configuration that still permits reconstruction. Specifically, having only 3 layer of points in the observation domain, i.e., x_W, x_I , and x_E where x_I is the physical boundary of both domains. To create a demanding test problem, the interface values imposed on the two independently solved subdomains were deliberately perturbed by a factor of 30. The reconstruction procedure developed in the previous sections was then applied using only the two overlap observations, x_W and x_E . Despite the extremely limited overlap width, the reconstruction successfully removed almost the entire interface-induced error. The corresponding error measures were

$$\|u_{\text{unc}} - u_{\text{exact}}\|_2 = 3.8681,$$

before reconstruction and

$$\|u_{\text{rec}} - u_{\text{exact}}\|_2 = 3.84 \times 10^{-2},$$

after reconstruction. This corresponds to a global improvement factor of approximately $100\times$. These values demonstrate that the dominant error introduced by the artificial interface conditions is accurately represented by the reconstructed homogeneous corrections even when the overlap consists of only a single physical node layer. The validity

Algorithm 1 Overlap-based reconstruction in one dimension

- 1: Partition the computational domain into overlapping subdomains Ω_W and Ω_E .
- 2: Solve Eq. (1) independently on each subdomain to obtain u_W and u_E .
- 3: Define the overlap region $\Omega_{\text{obs}} = [x_W, x_E]$.
- 4: Select two observation points $x_1, x_2 \in \Omega_{\text{obs}}$.
- 5: Compute the overlap mismatch

$$d_i = u_W(x_i) - u_E(x_i), \quad i = 1, 2.$$

- 6: Compute the decay parameter

$$\lambda = \sqrt{\frac{k}{h}}.$$

- 7: Construct the normalized basis functions $\phi_W(x)$ and $\phi_E(x)$ using Eqs. (4) and (5).
- 8: Assemble the observation vector $\mathbf{d}$ and reconstruction matrix $\mathbf{H}$ according to Eqs. (6) and (7).
- 9: Solve the reconstruction system

$$\mathbf{H}\mathbf{a} = \mathbf{d}$$

to obtain the amplitudes a_W and a_E .

- 10: Reconstruct the homogeneous corrections

$$u_h^W(x) = a_W \phi_W(x), \quad u_h^E(x) = a_E \phi_E(x).$$

- 11: Compute the corrected solutions

$$\tilde{u}_W = u_W - u_h^W, \quad \tilde{u}_E = u_E - u_h^E.$$

- 12: Define the weighting function

$$\omega(x) = \frac{x_E - x}{x_E - x_W}.$$

- 13: Construct the blended reconstruction

$$u_R(x) = \omega(x)\tilde{u}_W(x) + (1 - \omega(x))\tilde{u}_E(x).$$

- 14: Evaluate the residual

$$r(x) = u_W(x) - u_E(x) - (a_W \phi_W(x) - a_E \phi_E(x)),$$

and monitor $\|r\|_2$ or $\|r\|_\infty$ as an error indicator.

- 15: Use u_R as the reconstructed solution within the overlap region.
-

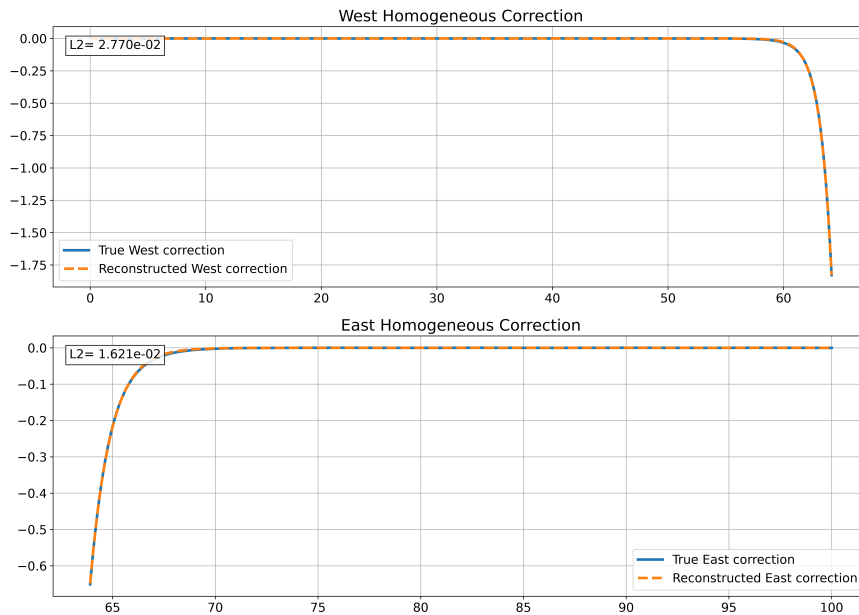


Figure 2: Comparison between the exact homogeneous corrections and the reconstructed homogeneous corrections for the minimal-overlap configuration $\delta_W = \delta_E = 1$. Even with only two overlap observations, the reconstructed corrections reproduce the interface-induced boundary-layer behaviour with high accuracy.

of the homogeneous reconstruction can be assessed by comparing the reconstructed corrections against the exact homogeneous errors obtained from the benchmark solution. Figures 2 and 3 shows the resulting comparison. For both subdomains the reconstruction error is approximately of the order of 2×10^{-2} , and the reconstructed profiles remain nearly indistinguishable from the exact homogeneous corrections throughout the computational domain, confirming that the boundary-layer structure generated by the artificial interface can be accurately recovered from only two observation points.

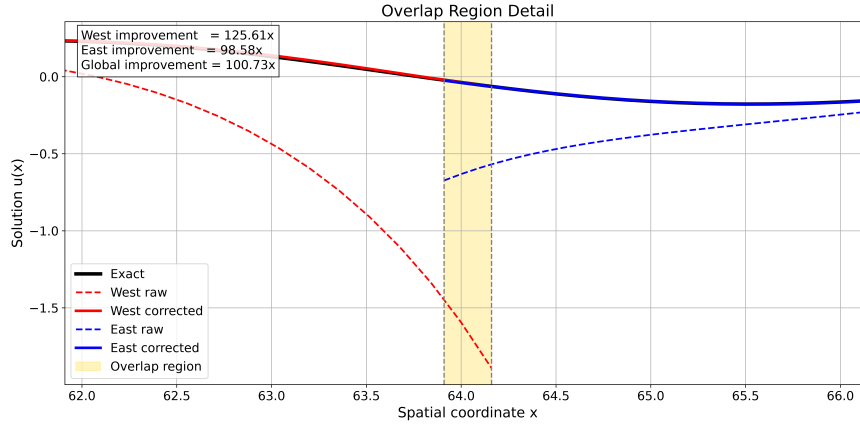


Figure 3: Detailed view of the overlap region for the minimal-overlap configuration. The dashed curves denote the independently computed subdomain solutions before reconstruction and the solid curves indicate the corrected solutions. Excellent agreement with the benchmark solution is obtained despite the extremely limited overlap width.

The excellent performance observed in Figures 2 and 3 is particularly significant because it demonstrates that overlap widths comparable to a ghost-node layer are sufficient for accurate reconstruction. This result suggests that the methodology can be incorporated directly into existing overlapping-grid implementations without requiring large overlap regions.

To assess the influence of mesh refinement, a convergence study was performed using

$$N = 50, 100, 200, 400, 800, 1600,$$

elements. All mesh levels were compared against a highly resolved reference solution computed using $N_{\text{ref}} = 10000$ elements. The resulting convergence behaviour is shown in Figure 4. The observed orders are

$$\begin{aligned} p_{\text{before}} &= 0.10, \\ p_{\text{sharp}} &= 0.89, \\ p_{\text{weighted}} &= 0.58. \end{aligned}$$

The uncorrected solution exhibits little convergence because the interface error dominates the numerical error. Both reconstructed solutions display substantially improved behaviour. Although classical finite-element convergence is not recovered completely, this is expected because the reconstruction employs only two overlap observations irrespective of mesh resolution. Consequently, the reconstruction eventually becomes information-limited rather than discretization-limited.

Overall, the results provide strong validation of the proposed methodology. Most importantly, they demonstrate that accurate reconstruction remains possible when the overlap width is reduced to a single physical node layer. From a practical perspective, this implies that interface information contained within ghost-node regions and their neighbouring physical nodes is sufficient to recover the dominant homogeneous correction. The ability to achieve global improvement factors exceeding two orders of magnitude using only two overlap observations strongly supports the feasibility of applying the proposed reconstruction framework within large-scale multidomain implementations of the Variational Boussinesq Model.

2.2 Two-Dimensional Problem

The reconstruction framework developed for the one-dimensional problem extends naturally to two spatial dimensions. The essential difference is that the homogeneous solution space is no longer spanned by a small number of exponentially decaying functions. Instead, interface errors may exhibit variation both normal and tangential to the artificial interface.

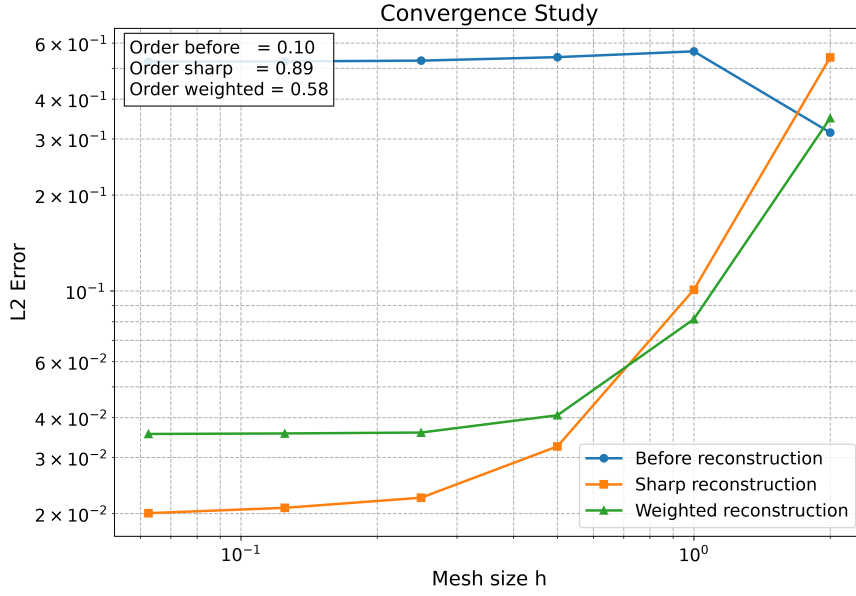


Figure 4: Mesh-refinement study for the minimal-overlap configuration. The reconstructed solutions converge substantially faster than the uncorrected solution. The observed error floor reflects the limited information content of the two-point reconstruction rather than instability of the reconstruction procedure.

Consequently, the homogeneous correction possesses a genuine modal structure and must generally be represented as a superposition of several independent modes.

The derivation of the two-dimensional homogeneous modes and the associated basis functions is presented in Appendix B. Only the resulting representation is summarized here. For a rectangular subdomain, the homogeneous correction is approximated by

$$\begin{aligned}
 u_h = & \sum_{n=1}^{N_m^W} a_n^{(W)} \phi_n^{(W)} + \sum_{n=1}^{N_m^E} a_n^{(E)} \phi_n^{(E)} \\
 & + \sum_{n=1}^{N_m^N} a_n^{(N)} \phi_n^{(N)} + \sum_{n=1}^{N_m^S} a_n^{(S)} \phi_n^{(S)},
 \end{aligned} \tag{9}$$

where $a_n^{(W)}$, $a_n^{(E)}$, $a_n^{(N)}$, and $a_n^{(S)}$ denote the modal amplitudes associated with the western, eastern, northern, and southern artificial interfaces, respectively. Unlike the one-dimensional setting, infinitely many admissible transverse modes exist. In practice only a finite number contribute significantly to the interface error and the expansion is truncated after N_m^j for $j = N, S, E, W$ modes. Figure 5 illustrates the first four homogeneous modes. As the mode number increases, additional oscillatory structure appears in the tangential direction, increasing the dimensionality of the reconstruction space.

Unlike the one-dimensional problem, the interface error can no longer be represented by a single decaying correction. Instead, different modes carry independent information about the interface perturbation. Consequently, reconstruction becomes a modal-amplitude recovery problem.

The overlap reconstruction strategy follows directly from the one-dimensional formulation. Consider two neighbouring overlapping subdomains, Ω_W, Ω_E . The subdomain decomposition introduce the interface Γ_I located at x_I . This interface can be interpreted, as before, the boundary between the two physical computational domains, while x_W and x_E are the location of the ghost-nodes respectively for the east domain and west domain. Unlike the one-dimensional problem, where the interface reduces to a single point, the interface in two dimensions is a curve embedded in the computational domain. As for the one-dimensional problem, the overlapping conditions induce a common region,

$$\Omega_{\text{obs}} = \Omega_W \cap \Omega_E,$$

which is referred to as the observation region. The geometry is illustrated schematically in Figure 6. A three-dimensional interpretation of the reconstruction process is shown in Figure 7.

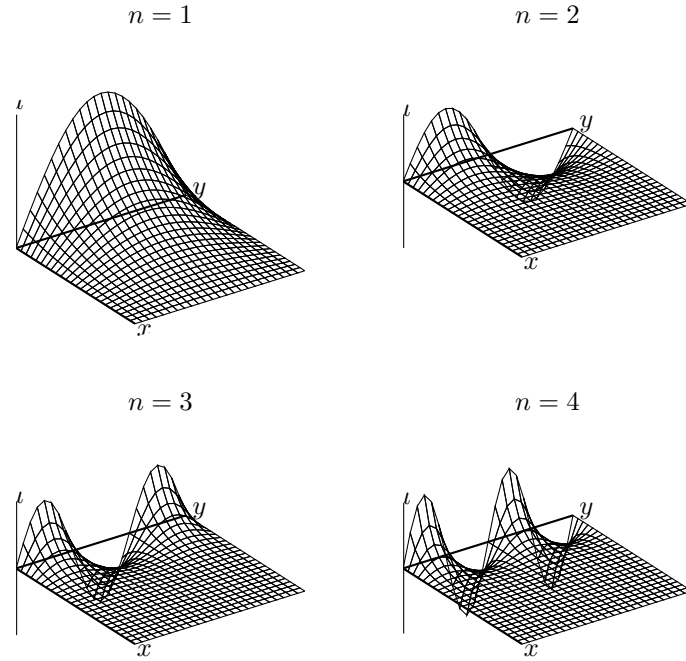


Figure 5: Examples of the first four homogeneous modes. Increasing mode number introduces additional tangential oscillations, thereby increasing the dimensionality of the reconstruction space.

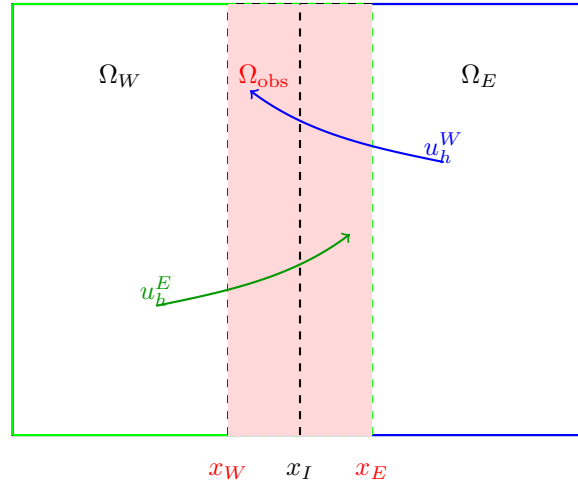


Figure 6: Overlapping-domain reconstruction in two dimensions. The overlap region Ω_{obs} contains observations from both subdomains. The homogeneous corrections u_h^W and u_h^E are associated with artificial interface conditions imposed at the overlap boundaries and are identified using information contained within the shared observation region.

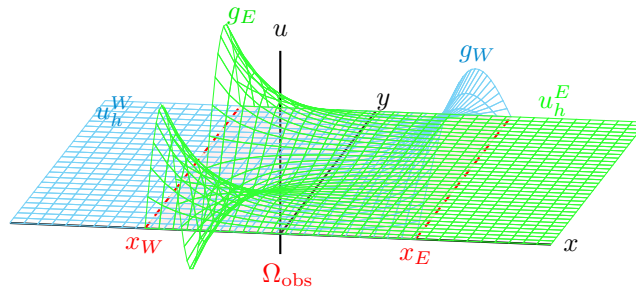


Figure 7: Three-dimensional interpretation of the overlap reconstruction mechanism. The overlap region Ω_{obs} is highlighted in red. The homogeneous corrections decay away from their respective artificial interfaces while coexisting throughout the observation region.

After solving the governing equation independently on both subdomains, the solutions may be decomposed as

$$u_W = u_p + u_h^W, \quad u_E = u_p + u_h^E.$$

Since both subdomains solve the same differential equation with identical forcing terms, the corresponding particular solutions are expected to coincide within the overlap region. Consequently,

$$u_W - u_E = u_h^W - u_h^E, \quad (x, y) \in \Omega_{\text{obs}}. \quad (10)$$

Using the modal representation Eq. (9), the homogeneous corrections become

$$u_h^W = \sum_j a_j^{(W)} \phi_j^{(W)}, \quad u_h^E = \sum_j a_j^{(E)} \phi_j^{(E)}.$$

Substituting into Eq. (10) gives

$$\begin{aligned} u_W(x_i, y_i) - u_E(x_i, y_i) &= \sum_j a_j^{(W)} \phi_j^{(W)}(x_i, y_i) \\ &\quad - \sum_j a_j^{(E)} \phi_j^{(E)}(x_i, y_i). \end{aligned} \quad (11)$$

Evaluating Eq. (11) at a collection of observation points produces the reconstruction system

$$\mathbf{H}\boldsymbol{\alpha} = \mathbf{d}, \quad (12)$$

where

$$d_i = u_W(x_i, y_i) - u_E(x_i, y_i),$$

and

$$\boldsymbol{\alpha} = \begin{bmatrix} a_1^{(W)} & \cdots & a_{N_m^W}^{(W)} & a_1^{(E)} & \cdots & a_{N_m^E}^{(E)} \end{bmatrix}^T.$$

The principal practical question is therefore not whether reconstruction is possible, but rather how much information must be extracted from the overlap region in order to recover the modal amplitudes accurately and efficiently. For a west-east decomposition with N_m^i for $i = W, E$ retained modes for each domains, the total number of unknown amplitudes is

$$N_{\text{unk}} = N_m^E + N_m^W.$$

Consequently, there is need to have at least N_{unk} of points. The square system is analogous to the one-dimensional formulation, although such a system is mathematically sufficient, it could be generally sensitive to perturbation in the numerical solution. In practical applications it could be advantageous to employ additional observations and solve the resulting system in a least-squares sense. Unlike the one-dimensional problem, however, the observations may be distributed in many different ways throughout the overlap region beside only along sample $x \in \Omega_{\text{obs}}$. Consequently, not only the number of observation points but also their spatial distribution influences the reconstruction accuracy. For a west-east decomposition, the simplest case is to collecting observations along two single sampling lines within the overlap region. For example,

$$x = x_W, \quad x = x_E,$$

or alternatively

$$x = x_W + \Delta_W, \quad x = x_E - \Delta_E.$$

In this configuration, observations are available at two distinct distances and therefore contain sufficient information to distinguish the decay behaviour associated with the western and eastern homogeneous corrections. The situation is directly analogous to the one-dimensional reconstruction. There, two independent observations were required to recover the two unknown homogeneous amplitudes a_W and a_E . In two dimensions, each retained mode introduces an independent amplitude associated with the corresponding domain implies an additional point along the sample lines.

The observations collected along each sampling line need not contain every available grid point. The reconstruction problem is fundamentally governed by the modal variation in the tangential direction. For a west-east decomposition, this variation occurs along y . Consequently, it is only necessary to resolve the tangential modal structure adequately. Once this structure is resolved, neighbouring observations become increasingly redundant and contribute little additional information to the reconstruction. The use of sampling lines, or any sampling method, yields a substantial reduction in computational cost. If every overlap cell is employed, the number of observations scales approximately as $\mathcal{O}(N_x N_y)$. By contrast, using only a small number of sampling lines reduces the observation count to at most $\mathcal{O}(N_y)$, if using all the points along the y -axis, which can be reduced to N_{unk} . For large overlap regions this reduction may significantly decrease the assembly and solution cost associated with the least-squares problem. The location of the sampling lines also influences the conditioning of the reconstruction matrix. Observations closely cluster together can produce basis evaluations that are highly correlated. The resulting reconstruction matrix may therefore become poorly conditioned. By contrast, observations distributed over multiple locations inside the overlap region tend to improve linear independence among the modal basis functions and lead to more robust amplitude estimates. In other words, clustered observations contain nearly redundant modal information. A useful strategy for constructing sparse observation sets is Farthest Point Sampling (FPS). Beginning from an initial observation point, FPS iteratively chooses the candidate point whose minimum distance to all previously selected points is maximal [2, 11]. The resulting distribution provides an approximately uniform spatial coverage of the overlap region while employing only a limited number of observations. Such space-filling strategies are particularly attractive because they maximize information content while minimizing redundancy [2, 11]. More generally, the same considerations apply to any method that seeks to select a sparse subset of observation points. Whether the observations are generated using FPS, low-discrepancy sequences, centroidal Voronoi distributions, greedy reduced-basis techniques, or other space-filling approaches, their effectiveness ultimately depends on the amount of geometric information available within the overlap region itself. In particular, when the distance between the overlap boundaries x_W and x_E becomes sufficiently small, the observation region degenerates into a narrow strip. In this regime, even an optimally distributed sparse point set may consist of observations that are located very close to each other. Consequently, the measurements contain increasingly redundant information and the rows of the reconstruction matrix become nearly linearly dependent. The reconstruction matrix $\mathbf{H}$ therefore becomes increasingly ill-conditioned. The issue is therefore not merely the number of observations but also the geometric diversity of the information they provide. As a consequence, increasing the number of sampled points does not necessarily improve reconstruction quality if those points remain confined to a very narrow overlap region. Once the overlap width becomes too small, all sparse sampling strategies naturally tend to generate highly correlated observations. This manifests itself through an increasing condition number $\kappa(\mathbf{H})$, greater sensitivity to numerical perturbations, and potentially larger errors in the recovered modal amplitudes. From a practical perspective, these observations suggest that the overlap width is at least as important as the observation strategy itself. Sparse sampling algorithms can only exploit the information available inside the overlap region; if the overlap region becomes excessively narrow, even an optimal choice of observation points cannot compensate for the lack of independent geometric information available to the reconstruction procedure.

An alternative reconstruction strategy is obtained by directly projecting the artificial boundary error onto the homogeneous modal basis. In this formulation, the boundary error is interpreted as a discrete distribution of source points located on the artificial interfaces and the modal amplitudes are obtained by projection rather than overlap matching. The corresponding derivation is presented in Appendix C.

2.2.1 Error Analysis

As in the one-dimensional formulation, the reconstruction is affected by discretization errors and numerical perturbations. Let

$$\begin{aligned} u_W &= u_p + u_h^W + \tau_W, \\ u_E &= u_p + u_h^E + \tau_E, \end{aligned}$$

where τ_W and τ_E denote the numerical errors associated with the independently computed subdomain solutions. Subtracting both expressions yields

$$u_W - u_E = u_h^W - u_h^E + \varepsilon,$$

with

$$\varepsilon = \tau_W - \tau_E.$$

The reconstruction system therefore becomes

$$\mathbf{H}\boldsymbol{\alpha} = \mathbf{d} + \boldsymbol{\varepsilon}.$$

Let $\boldsymbol{\alpha}^*$ denote the exact modal amplitudes and $\boldsymbol{\alpha}$ the reconstructed modal amplitudes. The reconstruction error is

$$\Delta\boldsymbol{\alpha} = \boldsymbol{\alpha} - \boldsymbol{\alpha}^*.$$

Using standard perturbation arguments,

$$\|\Delta\alpha\| \lesssim \kappa(\mathbf{H}) \|\varepsilon\|,$$

where $\kappa(\mathbf{H})$ denotes the condition number of the reconstruction matrix. Unlike the one-dimensional reconstruction, the multidimensional formulation introduces an additional source of error associated with observation placement. Two observation sets containing exactly the same number of points may produce reconstruction matrices with substantially different condition numbers. Poorly distributed observations may fail to capture the dominant modal structure and therefore lead to unstable amplitude estimates, even when the nominal number of observations is sufficient. The reconstruction accuracy therefore depends primarily on

1. the magnitude of the perturbation present in the overlap observations;
2. the conditioning of the reconstruction matrix, hence the spatial distribution of the observations;
3. the completeness of the retained modal basis.

The completeness of the retained modal basis represents a third source of reconstruction error. For rectangular subdomains, the homogeneous correction can be represented through separation-of-variables expansions. When contributions from all exterior boundaries are considered, the exact homogeneous correction may be written as in Equation 9. The numerical reconstruction retains only a finite number of modes so that the truncation error is

$$\begin{aligned} R_{N_m} = u_h - u_h^{(N_m)} &= \sum_{n=N_m+1}^{\infty} a_n^{(W)} \phi_n^{(W)} + \sum_{n=N_m+1}^{\infty} a_n^{(E)} \phi_n^{(E)} \\ &+ \sum_{n=N_m+1}^{\infty} a_n^{(N)} \phi_n^{(N)} + \sum_{n=N_m+1}^{\infty} a_n^{(S)} \phi_n^{(S)}. \end{aligned}$$

The origin of the truncation error can be understood from the classical separation-of-variables solution of elliptic problems. For Laplace-type equations in a rectangular region, the modal functions are products of trigonometric and hyperbolic functions, for example

$$\phi_n(x, y) = \sin\left(\frac{n\pi y}{H}\right) \sinh\left(\frac{n\pi x}{H}\right).$$

The corresponding modal amplitudes are determined by the Fourier expansion of the boundary data. Since higher-order modes are associated with increasingly large eigenvalues,

$$\lambda_n = \frac{n\pi}{H},$$

their influence decays rapidly away from the generating boundary. Consequently, for sufficiently smooth boundary data, the modal coefficients decrease rapidly with increasing mode number, exhibiting exponential decay of the form

$$|a_n| \lesssim C e^{-\beta n},$$

for some constant $\beta > 0$ depending on the regularity of the boundary data and the geometry of the overlap region. As a result,

$$\|R_{N_m}\| = \mathcal{O}(e^{-\beta N_m}),$$

and the truncation error decreases exponentially as additional modes are retained. This behaviour is consistent with the classical eigenfunction-expansion theory obtained through separation of variables for elliptic boundary-value problems (Ref. [13]). Increasing N_m therefore improves the approximation of the homogeneous correction by reducing the neglected modal content. However, it also increases the number of unknown amplitudes to be reconstructed and may lead to a more ill-conditioned least-squares problem. The choice of N_m therefore represents a compromise between truncation accuracy, numerical stability, and computational cost. Following the numerical perturbation reasoning, the corrected solutions $\tilde{u}$ are not expected to coincide exactly. To reduce the influence of these discrepancies and improve continuity throughout the overlap region, a weighted blending procedure can be used following ideas commonly employed in overlapping-grid and overset-grid methodologies [15, 1]. The reconstructed solution, as of the one-dimensional problem, is defined as

$$u_R(x, y) = \omega(x, y) \tilde{u}_W(x, y) + (1 - \omega(x, y)) \tilde{u}_E(x, y),$$

for $(x, y) \in \Omega_{\text{obs}}$, where $\omega(x, y) \in [0, 1]$. For a west-east decomposition, a natural partition-of-unity weighting is

$$\omega(x, y) = \frac{x_E - x}{x_E - x_W}.$$

This weighting gradually transfers the contribution from the western reconstruction to the eastern reconstruction across the overlap region. Similarly, for a north-south decomposition,

$$\omega(x, y) = \frac{y_N - y}{y_N - y_S}.$$

As in the one-dimensional reconstruction, this blending step acts as an additional averaging procedure that reduces the influence of local discretization errors and improves continuity throughout the overlap region. Similar blending approaches are widely employed in overset-grid and Chimera-grid methodologies [15, 1]. The complete reconstruction procedure is summarized in Algorithm 2.

Algorithm 2 Overlap reconstruction in two dimensions

- 1: Partition the computational domain into overlapping subdomains.
- 2: Solve Eq. (1) independently on each subdomain.
- 3: Define the overlap region

$$\Omega_{\text{obs}} = \Omega_W \cap \Omega_E.$$

- 4: Select at least two observation lines inside the overlap region.
- 5: Select observation points on the observation lines or throughout the overlap region.
- 6: Construct the homogeneous modal basis functions described in Appendix B.
- 7: Compute the overlap mismatch

$$d_i = u_W(x_i, y_i) - u_E(x_i, y_i).$$

- 8: Assemble the reconstruction matrix

$$H_{ij} = \phi_j(x_i, y_i).$$

- 9: Solve the least-squares problem

$$\boldsymbol{\alpha} = \arg \min_{\mathbf{z}} \|\mathbf{H}\mathbf{z} - \mathbf{d}\|_2.$$

- 10: Reconstruct the homogeneous corrections

$$u_h^W = \sum_j a_j^{(W)} \phi_j^{(W)},$$
$$u_h^E = \sum_j a_j^{(E)} \phi_j^{(E)}.$$

- 11: Compute

$$\tilde{u}_W = u_W - u_h^W, \quad \tilde{u}_E = u_E - u_h^E.$$

- 12: Construct the blending weight $\omega(x, y)$.

- 13: Form the reconstructed solution

$$u_R = \omega \tilde{u}_W + (1 - \omega) \tilde{u}_E.$$

- 14: Evaluate the residual

$$\mathbf{r} = \mathbf{d} - \mathbf{H}\boldsymbol{\alpha}.$$

- 15: Use u_R as the reconstructed solution within the overlap region.
-

2.2.2 2-Domains Test Case

As in the one-dimensional formulation, the proposed reconstruction methodology is first validated using a simple two-domain overlapping configuration consisting of a western and an eastern subdomain. The objective of this test case is to assess whether the modal reconstruction procedure is capable of recovering the homogeneous interface error introduced by an artificial domain decomposition while using only a very limited amount of information inside the overlap region. The computational domain is defined as $\Omega = [0, 100] \times [0, 40]$. The domain is partitioned into a western subdomain Ω_W and an eastern subdomain Ω_E , with the physical computational boundary for both domains at $x_I = 64$. To mimic the minimal-overlap configurations commonly encountered in the VBM-XMF framework, the overlap width is restricted to only a single layer on either side of the interface, meaning the observation domains has 3 layers x_W, x_I , and x_E . Consequently, the observation region consists of a narrow strip surrounding the interface. The governing equation is solved with $k = 25$, together with

$$\begin{aligned} h(x, y) &= 30, \\ p(x, y) &= 10 + 64 \cos(0.64x) \cos(0.25y), \\ f(x, y) &= \sin(0.25x) \sin(0.15y), \end{aligned}$$

and homogeneous Dirichlet boundary conditions imposed along all physical boundaries. A highly resolved reference solution is computed on the undecomposed domain using 400 and 600 finite elements along the x and y direction. The western and eastern subdomains are subsequently solved independently. Perturbed Dirichlet data are prescribed on the overlap boundaries x_W and x_E , which serve as the artificial boundaries of the eastern and western subdomain problems, respectively. The imposed boundary perturbations have the form

$$30 \sin\left(\frac{2\pi y}{L_y}\right),$$

thereby generating smooth interface errors with a non-trivial modal structure in the transverse direction. The homogeneous correction is approximated using 5 modes for each subdomain. The total number of unknown amplitudes is therefore 10. Following the sampling-line strategy introduced in the previous section, observations are collected only on the two overlap boundaries, x_W and x_E . Five observation points are distributed uniformly along each sampling line, yielding a total of 10 observations. Consequently, the resulting reconstruction system is square and contains exactly as many observations as unknown modal amplitudes.

Figure 8 shows the difference between the uncorrected overlapping-domain solution and the reference solution. Prior

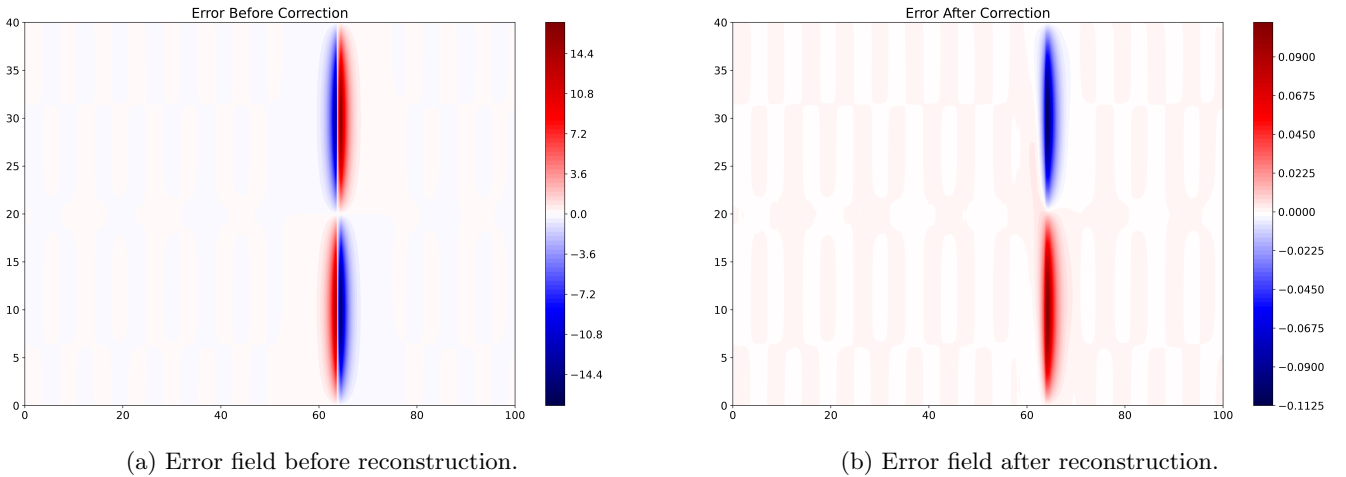


Figure 8: Comparison of the error field before and after reconstruction. Prior to reconstruction, the interface perturbations imposed at the artificial overlap boundaries generate a strongly localized modal error structure within the overlap region. After reconstruction, the dominant interface-induced modes are removed and only a small residual error remains near the interface. Despite employing only five observation points on each overlap boundary, for a total of ten observations, the reconstruction reduces the error by approximately two orders of magnitude.

to reconstruction, a pronounced error structure is visible near the artificial interface location $x_I = 64$. The error is strongly localized in the overlap region and exhibits a clear modal variation in the transverse direction. Maximum error magnitudes are approximately 1.5×10^1 , indicating that the imposed interface perturbation dominates the numerical error of the underlying discretization. After applying the modal reconstruction procedure, the dominant interface error is almost completely removed. The resulting error field is shown in Figure 8. The characteristic interface signature observed before reconstruction disappears almost entirely, leaving only a weak residual error concentrated near the interface. The maximum error is reduced from approximately 1.5×10^1 to 10^{-1} , corresponding to a reduction of roughly two orders of magnitude, or approximately 99% of the initial deviation. Despite the extremely sparse amount of information and the minimal overlap width, the reconstruction accurately recovers the dominant modal amplitudes associated with the homogeneous correction.

To investigate the influence of modal resolution and observation density, a convergence study was performed by simultaneously increasing the number of retained modes and the number of observations. For each experiment, N_m modes were retained on both subdomains and N_m observations were placed on each overlap boundary. Consequently,

the reconstruction system remained square throughout the study,

$$N_{\text{obs}} = N_{\text{unk}} = 2N_m.$$

The modal resolution and observation density were progressively increased until the sampling along the overlap boundaries approached the full transverse resolution available on the mesh. The resulting error and conditioning behaviour are shown in Figure 9. For very small values of N_m , the reconstruction error remains large. In this regime, the retained modal basis is unable to represent the dominant structure of the homogeneous correction generated by the artificial overlap boundary conditions. At the same time, only a limited number of observations is available to identify the unknown modal amplitudes, reducing the amount of information that can be extracted from the overlap region. The

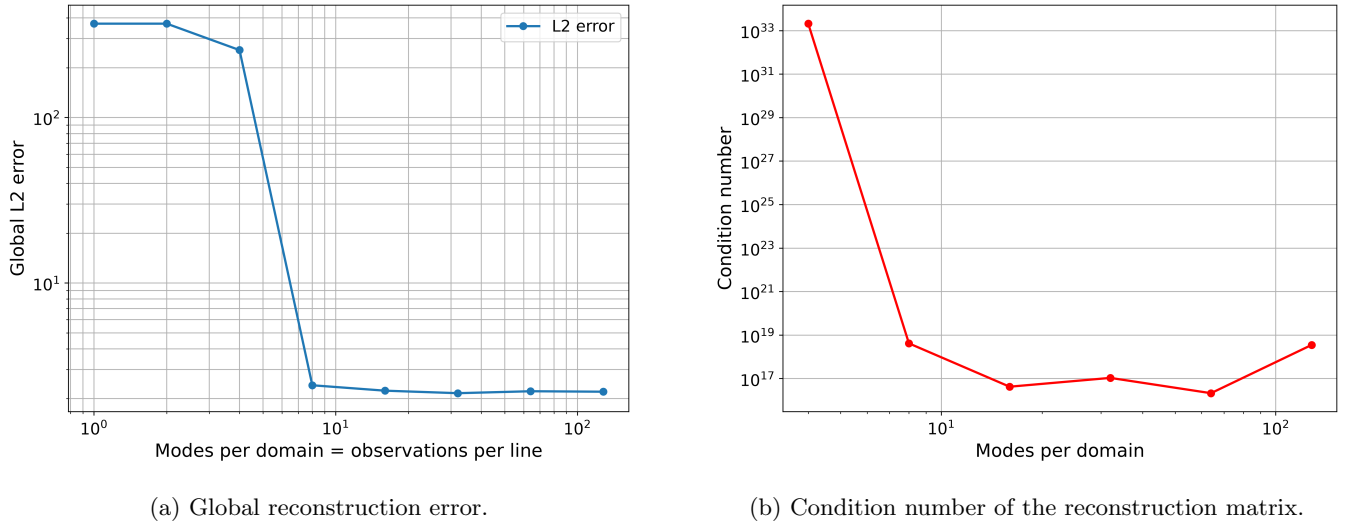


Figure 9: Influence of modal resolution and observation density on the reconstruction procedure. The number of observation points on each sampling line is chosen equal to the number of retained modes per domain, resulting in a square reconstruction system $N_{\text{obs}} = N_{\text{unk}} = 2N_m$.

imposed perturbation is proportional to

$$\sin\left(\frac{2\pi y}{L_y}\right),$$

and is therefore dominated by the second transverse mode. This behaviour is immediately reflected in the conditioning curve. When only one mode per domain is retained, the reconstruction space does not contain the dominant perturbation mode and the resulting reconstruction matrix is nearly singular. Increasing the modal resolution to two modes per domain introduces the dominant transverse structure into the reconstruction space and reduces the condition number by many orders of magnitude. The error behaviour shown in Figure 9a reveals that the reconstruction accuracy is not governed solely by the dominant perturbation mode. Although the second transverse mode is already available when $N_m = 2$, the reconstruction error remains large for $N_m = 1, 2$, and 4. This indicates that the homogeneous correction generated by the elliptic problem contains a richer modal structure than the imposed boundary perturbation itself. The solution of the elliptic boundary-value problem introduces additional modal components whose contribution must also be recovered by the reconstruction procedure. A particularly significant reduction in error occurs when increasing the resolution from four to eight modes per domain. In this range, two effects occur simultaneously. First, the modal basis becomes rich enough to represent a much larger portion of the homogeneous correction. Second, the number of observations increases from four to eight points on each overlap boundary, providing a substantially better description of the variation of the solution in the transverse direction. The combined effect produces the abrupt drop in error visible in Figure 9a. Beyond approximately eight modes per domain, only marginal improvements are observed and the error rapidly approaches a plateau. This suggests that the dominant homogeneous structure has already been captured and that further enrichment of the modal basis contributes little additional accuracy. At this stage, reconstruction accuracy is no longer controlled primarily by modal truncation. The conditioning results show that modal completeness is only one part of the reconstruction problem. Even around $N_m = 5$, corresponding to five observations on each overlap boundary and ten unknown modal amplitudes in total, the reconstruction matrix remains highly ill-conditioned. Consequently, the reconstruction error cannot be interpreted purely as a truncation error. Although the modal basis already contains the dominant components of the homogeneous correction, the corresponding amplitudes remain difficult to determine accurately because the inverse problem itself is poorly conditioned. This behaviour is consistent with the error analysis presented earlier. Increasing N_m enlarges the approximation space and reduces

modal truncation error, but accurate reconstruction additionally requires that the observations provide sufficiently independent information. In the present test case, the overlap consists of only a one-cell extension on either side of the interface. As a result, the two sampling lines $x = x_W$ and $x = x_E$ remain extremely close to one another. Although the number of observations increases, the geometric diversity of the measurements remains limited and the rows of the reconstruction matrix remain strongly correlated. The convergence study therefore demonstrates that reconstruction accuracy depends on two competing factors. The first is the completeness of the retained modal basis, which determines how accurately the homogeneous correction can be represented. The second is the conditioning of the reconstruction matrix, which determines how accurately the modal amplitudes can be recovered from the available observations. Increasing the number of modes alone is insufficient if the corresponding amplitudes cannot be identified robustly from the overlap data. Overall, the results show that the dominant interface-induced error can be recovered using a relatively small number of modes and observations. For the present test problem, fewer than ten observation points per sampling line already capture most of the homogeneous structure, while further increases in modal resolution provide only limited additional benefit. This confirms that the reconstruction error is governed not only by modal truncation but also by the conditioning of the underlying inverse problem. To investigate the influence of the overlap width independently of the modal resolution, an additional convergence study was performed while keeping both the modal basis and the observation strategy fixed. In all experiments, five modes were retained on each subdomain and five observations were placed on each overlap boundary, resulting in a reconstruction system containing ten unknown modal amplitudes and ten observations. Unlike the previous study, the number of modes and observation points remained unchanged. Instead, the overlap width was progressively increased by moving the observation boundaries x_W and x_E farther away from the interface location x_I . The resulting error behaviour is shown in Figure 10. A clear reduction in the global reconstruction error is observed as the overlap width increases. Starting from the minimal-overlap configuration, the error decreases by more than an order of magnitude before eventually reaching a plateau for larger overlaps. The significance of this result is that the modal basis is identical for every experiment. Consequently,

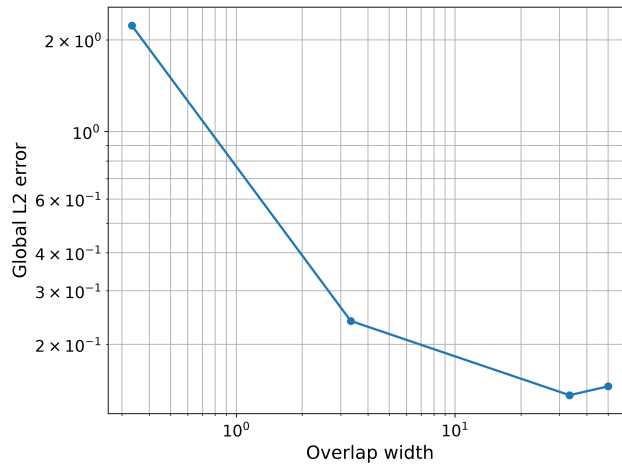


Figure 10: Influence of overlap width on reconstruction accuracy. The number of retained modes and observations remains fixed throughout the study. The observed improvement is therefore solely attributable to the increased distance between the observation boundaries x_W and x_E .

the reduction in error cannot be attributed to the availability of additional modes. Instead, it reflects an improvement in the quality of the information used to recover the modal amplitudes. When the overlap is extremely narrow, the two observation lines are located very close to one another. The resulting measurements therefore contain highly correlated information and provide only limited insight into the decay behaviour of the homogeneous corrections across the overlap region. Increasing the overlap width improves this situation. The observation lines become more widely separated and consequently sample the homogeneous correction at substantially different distances from the artificial boundaries. The reconstruction matrix therefore contains more independent information regarding the decay of the modal fields, leading to a more accurate recovery of the modal amplitudes. This interpretation is fully consistent with the theoretical error analysis presented earlier, which identified the geometric diversity of the observations as a key factor governing reconstruction accuracy. An additional observation is that the improvement saturates for sufficiently large overlaps. Beyond a certain width, the dominant modal structure has already been identified accurately and further increases in overlap provide only a modest reduction in error. In this regime, the remaining error is primarily associated with numerical discretization errors, the finite-dimensional modal approximation, and the intrinsic sensitivity of the amplitude-recovery problem. The overlap-width study therefore complements the modal convergence study. Whereas the previous experiment demonstrated the importance of a sufficiently rich modal basis, the present results

show that modal completeness alone is insufficient. Even when the modal basis is fixed, significant accuracy gains can be achieved by increasing the informational content of the overlap observations. Together, both studies confirm that successful reconstruction requires a balance between modal representation and observation quality.

2.2.3 4-Domains Test Case

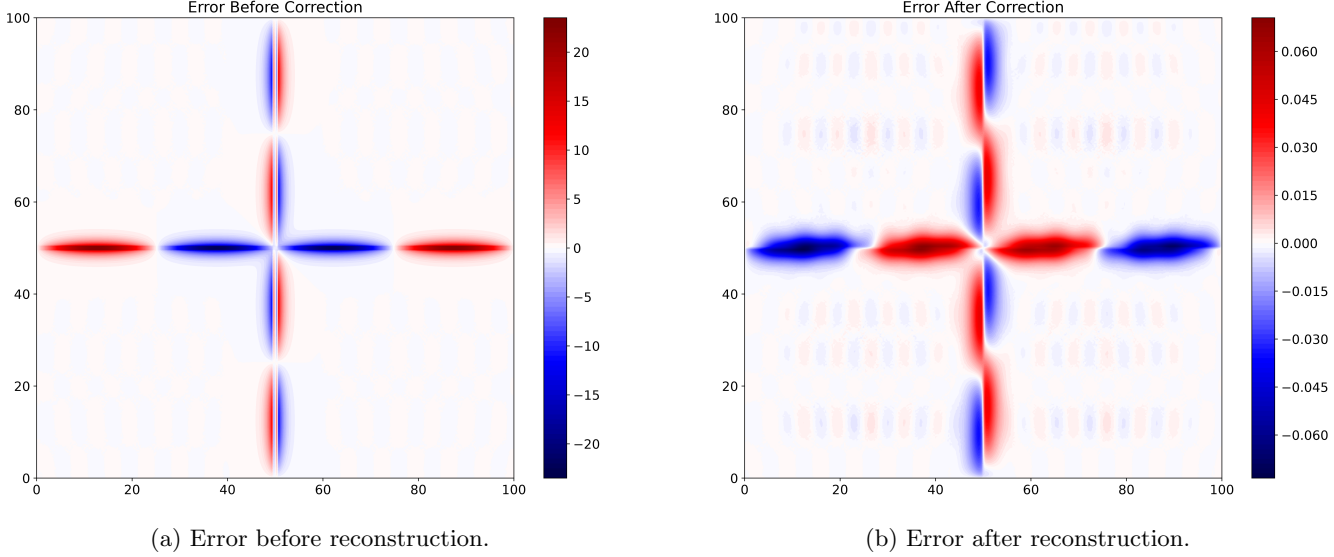


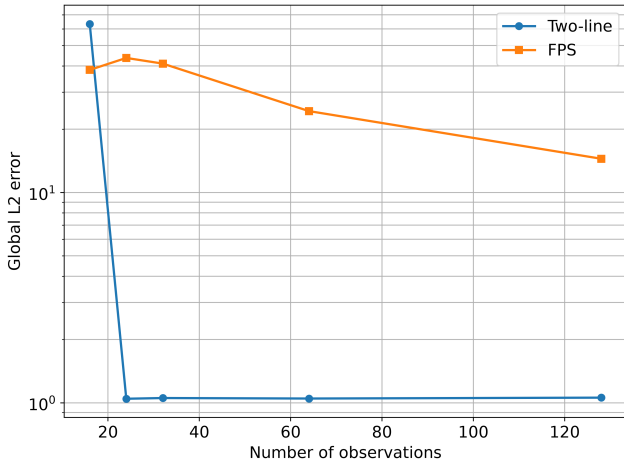
Figure 11: Four-domain reconstruction experiment. Before reconstruction, interface-induced errors are strongly localized along the horizontal and vertical decomposition interfaces and attain magnitudes of approximately 2.4×10^1 . After reconstruction, the dominant modal components are removed and only a small residual remains, with maximum errors of approximately 7×10^{-2} . The resulting reduction corresponds to more than two orders of magnitude improvement despite the minimal overlap width and the use of only sixteen unknown modal amplitudes reconstructed from thirty-two observations.

The final validation case considers the two-dimensional four-domain configuration introduced in Figure 6. Unlike the previous west-east decomposition, the computational domain is now partitioned simultaneously in both coordinate directions, producing four overlapping subdomains. The decomposition introduces both a vertical and a horizontal interface where the two physical computational domains coincide located at x_I , and y_I . Consequently, each subdomain possesses two artificial boundaries. For example, the south-west domain is bounded by an eastern interface and a northern interface, while the north-east domain possesses a western and a southern interface. The overlap region is again chosen as small as possible. Along the west-east decomposition only three layers are retained, x_W, x_I, x_E , while along the south-north decomposition the overlap consists of² y_S, y_I, y_N . This configuration closely resembles the ghost-node structure used within the VBM-XMF framework and represents an intentionally demanding reconstruction setting, in which x_W, x_E and y_N, y_S are the boundaries of the domains. The governing equation, boundary conditions, and perturbations are identical to the one employed in the previous test cases. Independent solutions are computed on all four subdomains using perturbed interface data. The imposed interface perturbations generate artificial homogeneous corrections attached to the overlap boundaries. These corrections propagate away from the interfaces and interact near the central overlap region where the horizontal and vertical decompositions intersect. Unlike the two-domain problem, each domain now contributes homogeneous corrections associated with two different interface directions. Using the modal representation introduced in Eq. (9), two modes are retained for each artificial interface. Since every domain contains two interfaces, the total number of unknown modal amplitudes becomes 16. Observations are collected on the overlap boundaries associated with both decompositions. Two observation locations are employed for each boundary and for each retained mode, resulting in a total of 32 observation points. Consequently, the reconstruction system is no longer square but overdetermined, and the modal amplitudes are obtained through a least-squares minimization of Eq. (12). Figure 11 shows the reconstruction error before and after applying the multidomain correction procedure. Prior to reconstruction, the dominant error is concentrated along the artificial interfaces. Strong vertical and horizontal error structures are visible near the shared boundaries. These structures correspond directly to the homogeneous corrections generated by the perturbed overlap boundary conditions. The largest errors occur near the intersection of the two interfaces, where the interface perturbations interact. Maximum error magnitudes approach 2.4×10^1 , indicating that the artificial interface conditions dominate the global solution error. After reconstruction, the interface-induced error is almost completely removed. As shown in Figure 11(b), only

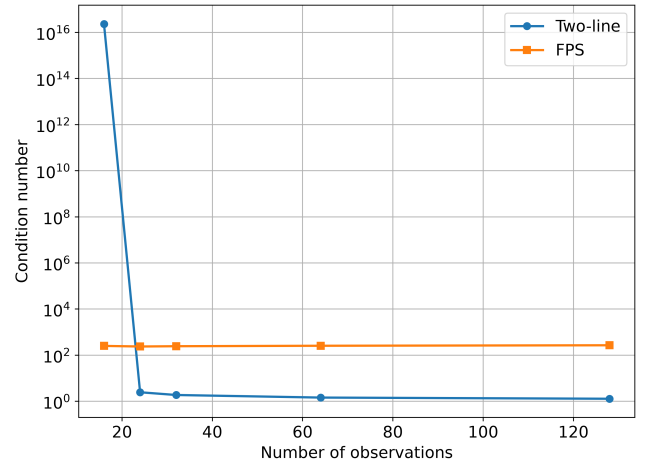
²Where S, and N stand for South and North.

a weak residual pattern remains near the interface intersection. The maximum error is reduced to approximately 7×10^{-2} . This corresponds to a reduction factor of approximately 3.4×10^2 , representing more than two orders of magnitude improvement. The most significant aspect of this result is that the reconstruction remains highly effective despite the extremely limited overlap width. The observation region contains only the three-layer structure x_W, x_I, x_E and y_S, y_I, y_N , yet the information available within these narrow overlap regions is sufficient to accurately identify all sixteen modal amplitudes. The results therefore confirm that the multidomain reconstruction procedure scales naturally from the two-domain setting to more complex decompositions. Even when multiple interfaces intersect and several families of homogeneous modes coexist simultaneously, the dominant interface-induced error can still be recovered from a relatively small number of observations. This strongly supports the feasibility of the approach for large multidomain decompositions encountered in practical VBM simulations.

The previous experiment employed a total of 32 observations despite the fact that only 16 unknown modal amplitudes were present in the reconstruction problem. At first sight, this choice may appear unnecessary, since a square system could already be constructed using 16 unknowns and observations. However, practical reconstruction accuracy is governed not only by the number of equations but also by the conditioning of the resulting inverse problem. When the overlap region is extremely narrow, as in the present configuration, the observations collected on neighbouring overlap boundaries contain highly correlated information. Although a square reconstruction system is mathematically sufficient, the resulting matrix may become nearly singular and therefore highly sensitive to numerical perturbations. To investigate this issue, an additional convergence study was performed in which the modal basis was kept fixed while the number of observations was progressively increased. Throughout the study, two modes were retained on each interface. Since each of the four subdomains possesses two artificial interfaces, the total number of unknown modal amplitudes remained fixed at 16. Only the observation density was varied. Two different observation strategies were considered. The first employs the structured sampling-line approach introduced previously. Two observation lines are placed on each overlap boundary and the number of sampling points along these lines is increased progressively. The second strategy employs Farthest Point Sampling (FPS), in which observation points are distributed throughout the overlap region according to a space-filling criterion. Figure 12 summarizes the resulting error and conditioning behaviour. Several observations can immediately be made. When only sixteen observations are employed, the two-line



(a) Global reconstruction error.



(b) Condition number of the reconstruction matrix.

Figure 12: Influence of the number of observations for the four-domain reconstruction problem. The modal basis remains fixed at sixteen unknown amplitudes while the observation density is increased.

strategy produces an extremely ill-conditioned reconstruction matrix. The condition number approaches

$$\kappa(\mathbf{H}) \approx 10^{16},$$

which is effectively at the limit of double-precision arithmetic. Consequently, the recovered modal amplitudes are highly sensitive to small perturbations in the overlap observations, resulting in a global reconstruction error of approximately

$$\|e\|_2 \approx 4 \times 10^1.$$

The behaviour changes dramatically once additional observations are introduced. Increasing the observation count from sixteen to twenty-four causes the condition number to collapse by more than fifteen orders of magnitude, approaching values close to unity. Simultaneously, the global reconstruction error drops from roughly 4×10^1 to values of

order unity. Beyond approximately twenty-four observations, both the error and the conditioning become essentially independent of the number of additional observations. This indicates that the reconstruction is no longer observation-limited and that the dominant homogeneous correction has already been identified accurately. The FPS strategy exhibits a substantially different behaviour. Throughout the study, the condition number remains relatively stable and of moderate magnitude, typically around $10^2 - 10^3$. This behaviour is expected since FPS distributes observations throughout the overlap region and therefore avoids the strong geometric correlation associated with observations confined to a small number of sampling lines. The resulting reconstruction matrices are therefore better conditioned even when relatively few observations are employed. Despite the improved conditioning, the reconstruction error obtained with FPS remains noticeably larger than for the two-line strategy. Even for large numbers of observations, the error remains approximately one order of magnitude larger than the corresponding line-sampling reconstruction. Nevertheless, FPS still achieves a substantial error reduction. Comparing the reconstructed solutions with the uncorrected solution shows that approximately two orders of magnitude of improvement are obtained throughout the entire study. This behaviour highlights an important distinction between conditioning and reconstruction accuracy. A well-conditioned reconstruction problem does not necessarily imply a more accurate reconstruction. The structured sampling-line strategy exploits prior knowledge regarding the analytical structure of the homogeneous modes. Since the dominant modal variation occurs along the overlap boundaries themselves, observations collected on these boundaries contain highly relevant information for recovering the modal amplitudes. FPS distributes points more uniformly throughout the overlap region, improving conditioning but also sampling regions that contain less direct information about the interface corrections. Consequently, the resulting amplitude estimates are more robust but generally less accurate. The previous study further suggested that overlap width itself may play an equally important role as the observation strategy. In particular, the width of the overlap region determines how much independent information is available regarding the decay of the homogeneous corrections away from the artificial boundaries. From the boundary-layer analysis presented earlier, the characteristic decay length of the homogeneous correction was found to be approximately four grid spacings. Since the four-domain configuration contains corrections generated from both sides of each overlap region, an overlap width of approximately $8 \times$ the grid size was selected for the FPS reconstructions. This choice ensures that the dominant decay behaviour associated with both overlapping boundary layers is contained within the observation region. To investigate the influence of overlap width, a second convergence study was performed in which the number of modes and observations were kept fixed while the overlap width was progressively increased. The resulting error behaviour is shown in Figure 13. The results reveal a clear reduction in the reconstruction error as the

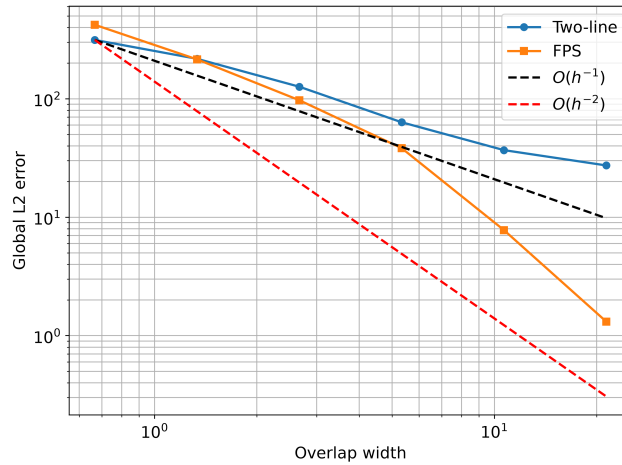


Figure 13: Influence of overlap width for the four-domain reconstruction problem. The modal basis and number of observations remain fixed throughout the study.

overlap width increases. For both sampling strategies, larger overlaps provide observations at increasingly different distances from the artificial interfaces and therefore contain more independent information regarding the decay of the homogeneous correction. The reconstruction matrix consequently becomes less sensitive to numerical perturbations and the recovered modal amplitudes become more accurate. Interestingly, the FPS strategy benefits particularly strongly from larger overlap widths. For narrow overlaps, the geometric diversity of the candidate observation cloud is limited and FPS cannot fully exploit its space-filling properties. As the overlap region expands, however, the candidate points become increasingly dispersed and FPS is able to generate substantially more informative observation sets. The resulting error decreases rapidly and eventually becomes significantly smaller than that obtained using the line-based strategy. These results confirm the conclusions of the theoretical error analysis. Successful reconstruction is governed by three competing factors:

1. the completeness of the retained modal basis;
2. the conditioning of the reconstruction matrix;
3. the geometric information content available within the overlap region.

Increasing the number of observations improves robustness, increasing the overlap width improves information content, and increasing the modal resolution reduces truncation error. Accurate reconstruction therefore requires a balance between all three effects rather than optimisation of only a single parameter.

2.3 Conclusion and Remarks

The results presented throughout this section provide strong evidence that the overlap region itself contains sufficient information to reconstruct the dominant interface-induced errors generated by artificial domain decomposition. The central idea underlying the proposed methodology is that the difference between independently computed overlapping subdomain solutions is primarily associated with homogeneous corrections. By identifying these corrections within the overlap region and removing them from the local solutions, it is possible to recover an approximation of the solution that would have been obtained on the original undecomposed domain without performing additional global elliptic solves. For the one-dimensional problem, the reconstruction framework admits a particularly simple representation. The homogeneous correction is described by two exponentially decaying modes whose amplitudes can be recovered directly from overlap observations. The numerical experiments demonstrated that even the smallest overlap configuration, consisting only of a single physical-node layer on either side of the interface, provides sufficient information for accurate reconstruction. Despite employing only two observation points, the reconstruction reduced the global error by approximately two orders of magnitude and recovered the dominant boundary-layer structure with high accuracy. These results indicate that overlap widths comparable to ghost-node layers are already sufficient for practical reconstruction. The extension to two spatial dimensions introduces a fundamentally different reconstruction problem. Instead of a small number of exponential corrections, the homogeneous solution possesses a genuine modal structure arising from separation-of-variables expansions of the underlying elliptic operator. The reconstruction therefore becomes a modal-amplitude recovery problem in which the accuracy depends on both the completeness of the retained basis and the quality of the available observations. The two-domain experiments demonstrated that the dominant interface perturbations can be represented using only a modest number of modes³. In the present test cases, fewer than ten modes per subdomain were sufficient to recover most of the homogeneous structure. The modal convergence studies further revealed that reconstruction accuracy is governed by more than modal completeness alone. Increasing the number of retained modes improves the approximation of the homogeneous correction through the exponential decay of the neglected modal contributions. However, the amplitudes associated with these modes must also be determined accurately from the overlap observations. Consequently, the reconstruction quality depends simultaneously on modal truncation error and on the conditioning of the inverse problem. A key observation throughout the two-dimensional studies is that the overlap width itself plays a central role in determining the quality of the reconstruction. When the overlap region becomes very narrow, the available observations are collected at nearly identical distances from the artificial interfaces. The resulting measurements therefore contain highly correlated information, producing poorly conditioned reconstruction matrices and large sensitivity to numerical perturbations. Increasing the overlap width improves the geometric diversity of the observations and substantially improves the recovery of the modal amplitudes. The overlap-width studies consistently showed a reduction in reconstruction error as the observation boundaries were moved farther away from the interface location. The four-domain experiments confirmed that the methodology extends naturally to multidomain configurations. Even when several families of homogeneous modes coexist simultaneously and multiple artificial interfaces intersect, the dominant interface-induced error can still be identified from local overlap observations. In the most demanding test case, sixteen unknown modal amplitudes were recovered successfully from a small set of overlap observations and produced global error reductions exceeding two orders of magnitude. The resulting improvement demonstrates that the reconstruction process scales beyond simple two-domain configurations and remains effective in multidomain environments that more closely resemble realistic VBM applications. The comparison between structured sampling lines and Farthest Point Sampling provides further insight into the nature of the reconstruction problem. Sampling along carefully selected overlap boundaries generally produces the most accurate reconstructions because the observations are located precisely where the modal information associated with the artificial interfaces is strongest. In contrast, FPS often produces better-conditioned systems by distributing observations more uniformly throughout the overlap region. However, improved conditioning does not necessarily imply improved reconstruction accuracy. The numerical experiments demonstrate that the most informative observations are not always the most uniformly distributed ones. Consequently, reconstruction performance is governed by the amount of physically relevant modal information captured by the observations rather than by conditioning alone. The theoretical and numerical investigations presented in this section therefore suggest that successful overlap reconstruction is controlled by three interacting mechanisms:

1. the completeness of the retained modal basis;

³Depending on the shape of the error.

2. the conditioning of the reconstruction matrix;
3. the geometric information content of the overlap region.

Increasing the number of modes reduces modal truncation error, increasing the number of observations improves robustness, and increasing the overlap width enriches the information available to the inverse problem. None of these factors alone is sufficient to guarantee accurate reconstruction. Instead, effective reconstruction requires a balance between representation quality, numerical stability, and observation diversity. From the perspective of practical VBM implementations, the most important conclusion is that accurate recovery of interface-induced errors is possible using only local information already available within the overlap region. No iterative Schwarz iterations, global correction equations, or additional elliptic solves are required. Since overlap regions naturally exist within ghost-node-based overset-grid formulations, the proposed methodology appears particularly well suited for integration into multidomain VBM solvers. The observed reductions in global error together with the modest number of required observations suggest that overlap-based homogeneous reconstruction provides a promising mechanism for coupling independently solved subdomains while maintaining high solution accuracy and low computational cost.

3 Disjoint Domains

Unlike the overlapping-domain formulation presented in the previous section, a disjoint domain decomposition contains no common observation region. Consequently, neighbouring subdomain solutions cannot be compared directly inside the computational domain. The central idea of the present section is therefore different from the overlap reconstruction strategy. Rather than recovering the homogeneous correction from observations collected inside a shared region, the reconstruction is performed directly at the artificial interface separating neighbouring subdomains. As in the overlapping formulation, the methodology exploits the linearity of the governing elliptic operator. The local errors introduced by the artificial interface conditions are interpreted as homogeneous corrections superimposed on the incorrectly solved subdomain solutions. The objective of the reconstruction is therefore to identify these homogeneous corrections and remove them such that the resulting solution satisfies the same continuity conditions that would hold on the original undecomposed domain. For elliptic problems, the globally correct solution is characterized by two fundamental interface conditions. The solution itself must be continuous across the interface and the normal flux must also be continuous. In the present formulation these two conditions provide exactly the information required to determine the unknown homogeneous correction amplitudes. The resulting reconstruction procedure differs fundamentally from the overlap method. In the overlap formulation, the amplitudes were recovered from observations collected within an observation region and the reconstruction problem was formulated as an inverse problem. In the disjoint formulation no observation region is available. Instead, the unknown amplitudes are obtained directly from the jumps in the solution and flux at the interface. Consequently, the reconstruction reduces to the solution of a small linear system whose size is determined entirely by the number of homogeneous modes retained. The notation follows the North–South–East–West convention introduced in Section 2 and Appendix A. In one dimension, the computational domain is partitioned into a western subdomain Ω_W and an eastern subdomain Ω_E , separated by an artificial interface located at $x = x_I$.

3.1 One-Dimensional Problem

Consider the one-dimensional domain

$$\Omega = [x_0, x_N].$$

The domain is partitioned into two disjoint subdomains,

$$\Omega_W = [x_0, x_I], \quad \Omega_E = [x_I, x_N].$$

Unlike the overlapping decomposition,

$$\Omega_W \cap \Omega_E = \{x_I\},$$

so that no overlap region exists. The geometry is illustrated in Figure 14. After solving the governing equation independently on both subdomains, the resulting solutions may be decomposed as

$$\begin{aligned} u_W &= u_p + u_h^W, \\ u_E &= u_p + u_h^E, \end{aligned}$$

where u_p denotes the particular solution that would have been obtained on the undecomposed domain and u_h^W, u_h^E denote the homogeneous corrections generated by the artificial interface conditions. As in the overlap formulation, the homogeneous corrections satisfy

$$-(hu_h')' + ku_h = 0. \quad (13)$$

The objective is to determine these corrections and remove them from the local solutions. For constant coefficients, Eq. (13) reduces to

$$-hu_h'' + ku_h = 0.$$

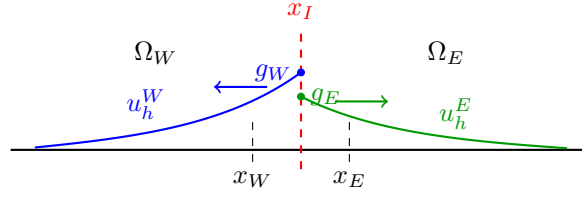


Figure 14: Disjoint-domain reconstruction. The computational domain is partitioned into a western subdomain Ω_W and an eastern subdomain Ω_E separated by the artificial interface x_I . Independent interface values g_W and g_E generate homogeneous corrections u_h^W and u_h^E that decay into their respective subdomains. Since no overlap region exists, the amplitudes of the homogeneous corrections are determined directly from continuity of the solution and continuity of the flux at the interface.

Seeking solutions of the form

$$u_h = e^{\lambda x}$$

gives

$$\lambda = \sqrt{\frac{k}{h}}.$$

As in the overlapping reconstruction, it is advantageous to construct normalized basis functions that automatically satisfy the physical boundary conditions. Define

$$\phi_W(x) = \frac{\sinh(\lambda(x - x_0))}{\sinh(\lambda(x_I - x_0))}, \quad (14)$$

and

$$\phi_E(x) = \frac{\sinh(\lambda(x_N - x))}{\sinh(\lambda(x_N - x_I))}, \quad (15)$$

which satisfy

$$\begin{aligned} \phi_W(x_0) &= 0, & \phi_W(x_I) &= 1, \\ \phi_E(x_N) &= 0, & \phi_E(x_I) &= 1. \end{aligned}$$

The homogeneous corrections are therefore represented as

$$u_h^W = a_W \phi_W, \quad (16)$$

$$u_h^E = a_E \phi_E, \quad (17)$$

where a_W and a_E denote the unknown correction amplitudes. The corrected solutions become

$$\tilde{u}_W = u_W - a_W \phi_W, \quad (18)$$

$$\tilde{u}_E = u_E - a_E \phi_E. \quad (19)$$

The unknown amplitudes are obtained from interface continuity.

The reconstructed solution must satisfy

$$\tilde{u}_W(x_I) = \tilde{u}_E(x_I). \quad (20)$$

Since

$$\phi_W(x_I) = \phi_E(x_I) = 1,$$

Eq. (20) becomes

$$u_W(x_I) - a_W = u_E(x_I) - a_E.$$

Introducing the interface solution jump

$$J_u = u_W(x_I) - u_E(x_I), \quad (21)$$

gives

$$a_W - a_E = J_u. \quad (22)$$

For elliptic problems, continuity of the solution alone is insufficient. The normal flux must also be continuous,

$$h(x_I) \tilde{u}'_W(x_I) = h(x_I) \tilde{u}'_E(x_I). \quad (23)$$

Define the interface fluxes

$$\begin{aligned} q_W &= h(x_I)u'_W(x_I), \\ q_E &= h(x_I)u'_E(x_I), \end{aligned}$$

and the associated flux jump

$$J_q = q_W - q_E. \quad (24)$$

Similarly, define

$$\begin{aligned} q_{\phi W} &= h(x_I)\phi'_W(x_I), \\ q_{\phi E} &= h(x_I)\phi'_E(x_I). \end{aligned}$$

Substituting the corrected solutions into Eq. (23) yields

$$q_{\phi W}a_W - q_{\phi E}a_E = J_q. \quad (25)$$

Combining Eqs. (22) and (25) gives

$$\begin{bmatrix} 1 & -1 \\ q_{\phi W} & -q_{\phi E} \end{bmatrix} \begin{bmatrix} a_W \\ a_E \end{bmatrix} = \begin{bmatrix} J_u \\ J_q \end{bmatrix} \quad (26)$$

for the unknown correction amplitudes.

Unlike the overlap reconstruction introduced in Section 2, the conditioning behaviour of the disjoint-domain formulation is fundamentally different. The overlap reconstruction matrix may become poorly conditioned when the overlap width becomes very small or when the observations contain highly correlated information. By contrast, the present reconstruction matrix is only

$$\mathbf{A} = \begin{bmatrix} 1 & -1 \\ q_{\phi W} & -q_{\phi E} \end{bmatrix},$$

and therefore contains only two unknown amplitudes. The principal question is whether

$$q_{\phi W} \approx q_{\phi E},$$

since this would cause the rows of the system to become nearly linearly dependent. For

$$\phi_W = \frac{\sinh(\lambda(x - x_0))}{\sinh(\lambda L_W)},$$

with

$$L_W = x_I - x_0,$$

one obtains

$$q_{\phi W} = h\lambda \coth(\lambda L_W).$$

Similarly,

$$\phi_E = \frac{\sinh(\lambda(x_N - x))}{\sinh(\lambda L_E)},$$

with

$$L_E = x_N - x_I,$$

gives

$$q_{\phi E} = -h\lambda \coth(\lambda L_E).$$

The determinant of the reconstruction matrix therefore becomes

$$\begin{aligned} \det(\mathbf{A}) &= q_{\phi W} - q_{\phi E} \\ &= h\lambda [\coth(\lambda L_W) + \coth(\lambda L_E)]. \end{aligned} \quad (27)$$

For large domains,

$$\coth(\lambda L) \rightarrow 1,$$

and therefore

$$\det(\mathbf{A}) \rightarrow 2h\lambda.$$

Consequently, the determinant remains bounded away from zero and the reconstruction matrix remains well conditioned even for very large domains. This behaviour differs significantly from the overlap reconstruction problem, where increasing domain size or reducing overlap width may cause severe conditioning issues. Thus, normalization of the coordinates may improve numerical scaling and interpretation, but it does not fundamentally alter the conditioning. The conditioning is governed primarily by the decay parameter $\lambda = \sqrt{k/h}$ rather than by the physical domain length itself.

3.1.1 Error Analysis

The previous derivation assumes exact subdomain solutions. In practice, discretization errors introduce perturbations in both the solution jump and the flux jump evaluated at the interface. Let

$$\begin{aligned} u_W &= u_p + u_h^W + \tau_W, \\ u_E &= u_p + u_h^E + \tau_E, \end{aligned}$$

where τ_W and τ_E denote the numerical discretization errors associated with the western and eastern subdomains. The computed solution jump becomes

$$J_u = (a_W - a_E) + \varepsilon_u,$$

where

$$\varepsilon_u = \tau_W(x_I) - \tau_E(x_I).$$

Similarly,

$$J_q = q_{\phi W} a_W - q_{\phi E} a_E + \varepsilon_q,$$

where

$$\varepsilon_q = h(x_I) [\tau'_W(x_I) - \tau'_E(x_I)].$$

The reconstruction system therefore becomes

$$\mathbf{A}\mathbf{a} = \mathbf{b} + \boldsymbol{\varepsilon},$$

with

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_u \\ \varepsilon_q \end{bmatrix}.$$

Let

$$\mathbf{a}^* = \begin{bmatrix} a_W^* \\ a_E^* \end{bmatrix}$$

denote the exact amplitudes and define

$$\Delta\mathbf{a} = \mathbf{a} - \mathbf{a}^*.$$

Then

$$\Delta\mathbf{a} = \mathbf{A}^{-1}\boldsymbol{\varepsilon}.$$

Consequently,

$$\|\Delta\mathbf{a}\| \leq \|\mathbf{A}^{-1}\| \|\boldsymbol{\varepsilon}\|.$$

Since the reconstruction matrix remains well conditioned, the reconstruction error is expected to be governed mainly by the interface perturbations themselves rather than by amplification due to the reconstruction procedure. For a second-order finite-difference discretization,

$$\tau_W, \tau_E = \mathcal{O}(\Delta x^2),$$

which implies

$$\varepsilon_u = \mathcal{O}(\Delta x^2).$$

Because the flux jump involves derivatives,

$$\varepsilon_q = \mathcal{O}(\Delta x).$$

The flux evaluation therefore frequently becomes the dominant source of reconstruction error. An a posteriori measure of reconstruction quality is provided by the residuals

$$r_u = \tilde{u}_W(x_I) - \tilde{u}_E(x_I),$$

and

$$r_q = h(x_I)\tilde{u}'_W(x_I) - h(x_I)\tilde{u}'_E(x_I).$$

Ideally,

$$r_u = 0, \quad r_q = 0.$$

Small residuals indicate that the homogeneous correction successfully removes the inconsistency generated by the artificial interface conditions, whereas large residuals may indicate inadequate spatial resolution, inaccurate flux evaluation, rapidly varying coefficients, or discretization errors that cannot be represented accurately by the retained homogeneous basis functions. The complete reconstruction procedure is summarized in Algorithm 3.

Algorithm 3 Disjoint-domain reconstruction in one dimension

- 1: Partition the domain into Ω_W and Ω_E .
- 2: Solve Eq. (1) independently on both subdomains.
- 3: Compute the local solutions u_W and u_E .
- 4: Compute

$$\lambda = \sqrt{\frac{k}{h}}.$$

- 5: Construct ϕ_W and ϕ_E .
- 6: Evaluate

$$J_u = u_W(x_I) - u_E(x_I).$$

- 7: Evaluate

$$J_q = h(x_I)u'_W(x_I) - h(x_I)u'_E(x_I).$$

- 8: Compute

$$q_{\phi W} = h(x_I)\phi'_W(x_I), \quad q_{\phi E} = h(x_I)\phi'_E(x_I).$$

- 9: Assemble and solve Eq. (26).

- 10: Compute

$$u_h^W = a_W\phi_W, \quad u_h^E = a_E\phi_E.$$

- 11: Reconstruct

$$u_R = \begin{cases} u_W - a_W\phi_W, & x \in \Omega_W, \\ u_E - a_E\phi_E, & x \in \Omega_E. \end{cases}$$

- 12: Evaluate the residuals r_u and r_q .
 - 13: Use u_R as the reconstructed solution.
-

3.1.2 2-Domains Test Case

The objective of the present experiment is to validate the flux-based reconstruction methodology developed in the previous section. In contrast to the overlap reconstruction procedure, no observation region is available and no inverse problem must be solved. The reconstruction relies solely on the continuity of the solution and the continuity of the flux at the artificial interface. The unknown correction amplitudes are therefore recovered directly from the interface jumps using the reconstruction system Eq. (26). The computational domain is defined as $\Omega = [0, 100]$. An artificial interface is introduced at $x_I = 64$, resulting in the disjoint decomposition $\Omega_W = [0, 64]$, $\Omega_E = [64, 100]$. The governing equation is solved with $k = 25$, together with the coefficient functions

$$\begin{aligned} h(x) &= 30 + 5 \cos(0.64x), \\ p(x) &= 10 + 64 \cos(0.64x), \\ f_x(x, t) &= 0.25 \cos(0.25x - 0.64t). \end{aligned}$$

A benchmark solution is first computed on the undecomposed domain using 800 finite elements. The western and eastern subdomains are subsequently solved independently using 400 finite elements in each subdomain. To create a challenging reconstruction problem, inconsistent interface values are prescribed on both sides of the interface. These perturbations generate homogeneous corrections attached to the artificial interface, producing both a solution discontinuity and a flux discontinuity. The reconstruction procedure introduced previously is then applied. The homogeneous corrections are represented using the normalized basis functions ϕ_W and ϕ_E , and the unknown amplitudes a_W and a_E are recovered from the continuity conditions imposed at the interface. Since only two amplitudes are present, the reconstruction reduces to the solution of a 2×2 linear system. Figure 15 compares the benchmark solution, the uncorrected multidomain solution, and the reconstructed solution. Prior to reconstruction, a large localized error is present at the artificial interface. The corresponding global error is

$$\|u_{\text{unc}} - u_{\text{exact}}\|_2 = 1.312 \times 10^1.$$

The error is dominated by the interface-generated homogeneous corrections and is strongly localized near $x_I = 64$. After reconstruction, the interface discontinuity is almost completely removed and the reconstructed solution becomes visually indistinguishable from the benchmark solution. The corresponding error decreases to

$$\|u_{\text{rec}} - u_{\text{exact}}\|_2 = 4.148 \times 10^{-3}.$$

This corresponds to an improvement factor of

$$\frac{\|u_{\text{unc}} - u_{\text{exact}}\|_2}{\|u_{\text{rec}} - u_{\text{exact}}\|_2} = 3.164 \times 10^3.$$

In other words, the reconstruction removes more than three orders of magnitude of the interface-induced error while employing only two homogeneous correction amplitudes. The quality of the reconstruction can be assessed more

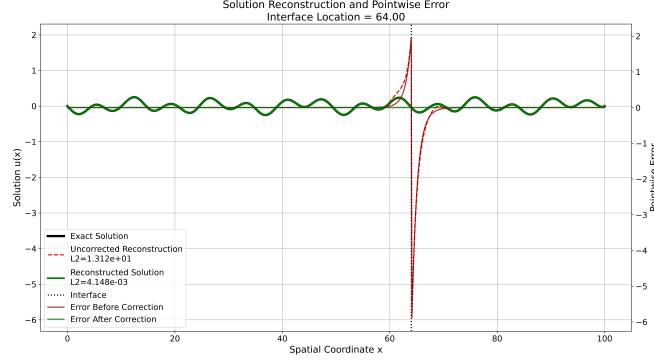


Figure 15: Comparison between the benchmark solution, the uncorrected disjoint-domain solution, and the reconstructed solution. Prior to reconstruction, the artificial interface introduces a pronounced localized error at $x_I = 64$. The flux-based reconstruction removes the interface inconsistency and reduces the global L_2 -error from 1.312×10^1 to 4.148×10^{-3} , corresponding to an improvement factor of approximately 3.16×10^3 .

directly by examining the homogeneous corrections themselves. Since the benchmark solution is available, the exact homogeneous corrections may be extracted by subtracting the benchmark solution from the independently computed subdomain solutions. Figure 16 compares these exact homogeneous corrections with the reconstructed corrections obtained from the interface reconstruction procedure. For the western subdomain, the reconstruction error is

$$\|u_h^W - \hat{u}_h^W\|_2 = 3.221 \times 10^{-3},$$

while for the eastern subdomain

$$\|u_h^E - \hat{u}_h^E\|_2 = 2.100 \times 10^{-3}.$$

The reconstructed corrections are almost indistinguishable from the exact homogeneous corrections throughout the entire computational domain. This confirms that the dominant error generated by the artificial interface is accurately represented by the homogeneous basis functions introduced in Eqs. (14) and (15). A second validation is obtained

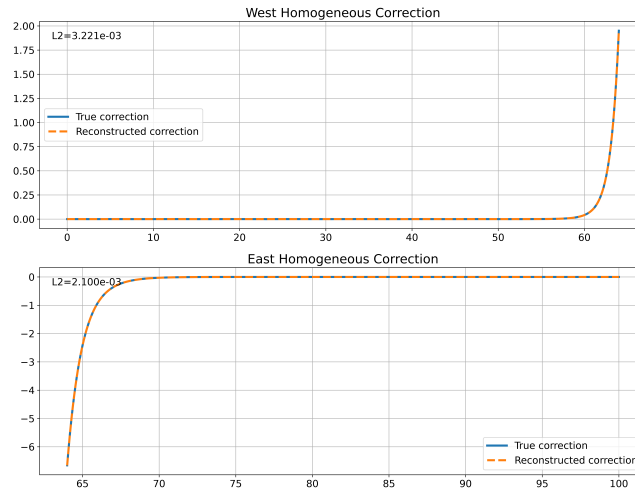


Figure 16: Comparison between the exact homogeneous corrections and the reconstructed homogeneous corrections. The upper panel shows the western homogeneous correction and the lower panel shows the eastern homogeneous correction. The reconstructed profiles remain nearly indistinguishable from the exact corrections, demonstrating that the interface-induced error is accurately recovered throughout both subdomains.

by directly examining the interface conditions after reconstruction. The theoretical development presented in the

previous subsection predicts that the solution jump and flux jump should vanish after application of the homogeneous correction. In the present experiment, the interface solution residual was found to be

$$r_u = u_{W,c}(x_I) - u_{E,c}(x_I) = -8.88 \times 10^{-16},$$

while the flux residual became

$$r_q = q_{W,c} - q_{E,c} = -1.68 \times 10^{-13}.$$

Both quantities are effectively at machine precision. Consequently, the reconstructed solution satisfies both continuity of the solution and continuity of the flux to numerical precision. This provides strong confirmation that the reconstructed homogeneous corrections successfully recover the interface information lost during the independent subdomain solves. An important distinction with respect to the overlap reconstruction concerns the conditioning of the reconstruction problem. In the overlap formulation, the observation matrix is assembled from basis-function evaluations in the overlap region and may become highly ill-conditioned when the overlap becomes small or the observations become nearly redundant. The disjoint formulation does not suffer from this difficulty. The reconstruction matrix remains

$$\mathbf{A} = \begin{bmatrix} 1 & -1 \\ q_{\phi W} & -q_{\phi E} \end{bmatrix},$$

which contains only two unknown amplitudes. For the present experiment, the condition number was

$$\kappa(\mathbf{A}) = 2.51 \times 10^1.$$

This value is modest and indicates that the reconstruction system is well conditioned. The condition number is many orders of magnitude smaller than those encountered in several of the overlap-reconstruction experiments presented in Section 2. Consequently, the present reconstruction is not especially sensitive to numerical perturbations in the interface data. The favourable conditioning observed numerically is consistent with the theoretical analysis developed in the previous subsection. The determinant of the reconstruction matrix is

$$\det(\mathbf{A}) = h\lambda [\coth(\lambda L_W) + \coth(\lambda L_E)],$$

which remains bounded away from zero even when the subdomains become very large. The conditioning of the reconstruction problem therefore depends primarily on the local properties of the differential operator rather than on the physical size of the computational domain. Overall, the results provide strong validation of the proposed flux-based reconstruction methodology. The homogeneous corrections are recovered with high accuracy, the solution and flux discontinuities are removed to machine precision, and the reconstructed solution closely matches the benchmark solution obtained on the undecomposed domain. Most importantly, these results are achieved without any overlap region, least-squares reconstruction, or inverse observation problem. The reconstruction relies exclusively on local interface information and therefore provides a natural framework for the treatment of disjoint domain decompositions.

3.2 Two-Dimensional Problem

The overlap reconstruction presented in Section 2 exploits the existence of an observation region in which neighbouring subdomain solutions coexist. The homogeneous correction amplitudes are recovered by comparing the local solutions inside this region. For non-overlapping decompositions, however, no common observation domain exists. Information can only be exchanged through the artificial interfaces separating neighbouring subdomains. Consequently, the reconstruction must be formulated entirely in terms of interface quantities. As in the one-dimensional formulation, the reconstruction is based on the observation that the error introduced by the artificial interface conditions belongs to the homogeneous solution space. Rather than recovering this error from overlap observations, the amplitudes are obtained by enforcing the physical interface conditions satisfied by the exact solution. For elliptic problems these consist of continuity of the solution and continuity of the normal flux across the interface. Consider two neighbouring subdomains Ω_W, Ω_E , sharing a common interface Γ_I . The notation follows the west–east convention introduced throughout this work. The same derivation applies to north–south interfaces after exchanging the coordinate directions. After independently solving Eq. (1) on both subdomains, the local solutions may be written as

$$\begin{aligned} u_W &= u_p + u_h^{(W)}, \\ u_E &= u_p + u_h^{(E)}, \end{aligned}$$

where u_p denotes the exact physical solution restricted to each subdomain and $u_h^{(W)}, u_h^{(E)}$ represent the homogeneous corrections generated by the artificial interface conditions. Since both subdomains solve the same governing equation with identical forcing terms, the particular solution is common to both domains. Consequently,

$$u_W - u_E = u_h^{(W)} - u_h^{(E)}, \quad (28)$$

so that the entire interface mismatch is contained within the homogeneous solution space.

3.2.1 Homogeneous Reconstruction Space

The homogeneous corrections are represented using the modal basis introduced in Appendix B. For a general rectangular subdomain, the homogeneous correction is approximated by

$$u_h = \sum_{n=1}^{N_m^W} a_n^{(W)} \phi_n^{(W)} + \sum_{n=1}^{N_m^E} a_n^{(E)} \phi_n^{(E)} + \sum_{n=1}^{N_m^N} a_n^{(N)} \phi_n^{(N)} + \sum_{n=1}^{N_m^S} a_n^{(S)} \phi_n^{(S)}. \quad (29)$$

Only the modal families associated with active artificial interfaces are retained. For example, if a subdomain possesses only an eastern artificial boundary,

$$u_h = \sum_{n=1}^{N_m^E} a_n^{(E)} \phi_n^{(E)}.$$

Similarly, if both eastern and northern interfaces are present,

$$u_h = \sum_{n=1}^{N_m^E} a_n^{(E)} \phi_n^{(E)} + \sum_{n=1}^{N_m^N} a_n^{(N)} \phi_n^{(N)}.$$

The reconstruction therefore adapts naturally to arbitrary multidomain configurations. Let

$$N_W = \dim(u_h^{(W)}),$$

and

$$N_E = \dim(u_h^{(E)}),$$

denote the number of unknown modal amplitudes retained on each side of the interface. Collecting all amplitudes yields

$$\boldsymbol{\alpha} = \begin{bmatrix} a_1^{(W)} & \dots & a_{N_W}^{(W)} & a_1^{(E)} & \dots & a_{N_E}^{(E)} \end{bmatrix}^T,$$

with

$$N_{\text{unk}} = N_W + N_E.$$

The reconstruction problem therefore consists of recovering the modal amplitude vector $\boldsymbol{\alpha}$.

3.2.2 Interface Continuity Conditions

The reconstructed solution must satisfy continuity of both the solution and the normal flux across the interface. The solution residual is defined by

$$R_u = u_W - u_E, \quad (30)$$

while the flux residual is

$$R_q = h_W \frac{\partial u_W}{\partial n} - h_E \frac{\partial u_E}{\partial n}, \quad (31)$$

where n denotes the interface normal. Substituting the homogeneous expansions gives

$$R_u = (u_W - u_E) + (u_h^{(W)} - u_h^{(E)}), \quad (32)$$

and

$$R_q = h_W \frac{\partial u_W}{\partial n} - h_E \frac{\partial u_E}{\partial n} + h_W \frac{\partial u_h^{(W)}}{\partial n} - h_E \frac{\partial u_h^{(E)}}{\partial n}. \quad (33)$$

The objective of the reconstruction is to enforce

$$R_u = 0, \quad R_q = 0,$$

everywhere along the interface.

3.2.3 Galerkin Reconstruction

As in finite-element formulations, the interface residuals are projected onto a set of test functions

$$\psi_i, \quad i = 1, \dots, N_m.$$

The weak continuity conditions are

$$\int_{\Gamma_I} \psi_i R_u d\Gamma = 0, \quad (34)$$

and

$$\int_{\Gamma_I} \psi_i R_q d\Gamma = 0. \quad (35)$$

Introducing

$$M_{ij} = \int_{\Gamma_I} \psi_i \phi_j d\Gamma, \quad (36)$$

and

$$K_{ij} = \int_{\Gamma_I} \psi_i h \frac{\partial \phi_j}{\partial n} d\Gamma, \quad (37)$$

the continuity and flux equations can be assembled into the global reconstruction system

$$\mathbf{A} \boldsymbol{\alpha} = \mathbf{b}, \quad (38)$$

where

$$\mathbf{A} = \begin{bmatrix} \mathbf{M}_W & -\mathbf{M}_E \\ \mathbf{K}_W & -\mathbf{K}_E \end{bmatrix},$$

and

$$\mathbf{b} = - \begin{bmatrix} \int_{\Gamma_I} \psi_i (u_W - u_E) d\Gamma \\ \int_{\Gamma_I} \psi_i \left(h_W \frac{\partial u_W}{\partial n} - h_E \frac{\partial u_E}{\partial n} \right) d\Gamma \end{bmatrix}.$$

Equation (38) constitutes the most general form of the non-overlapping reconstruction problem.

3.2.4 Modal Decoupling

For rectangular domains with locally constant coefficients, the homogeneous modes become orthogonal along the interface. For a vertical west–east interface,

$$\psi_n(y) = \sin \left(\frac{n\pi y}{L_y} \right),$$

and

$$\phi_n(x_I, y) \propto \sin \left(\frac{n\pi y}{L_y} \right).$$

The orthogonality relation

$$\int_0^{L_y} \sin \left(\frac{n\pi y}{L_y} \right) \sin \left(\frac{m\pi y}{L_y} \right) dy = 0, \quad n \neq m,$$

causes the off-diagonal entries of both $\mathbf{M}$ and $\mathbf{K}$ to vanish. In this ideal limit,

$$\mathbf{A} = \text{diag}(\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_{N_m}),$$

and the reconstruction separates into independent modal systems. The one-dimensional flux reconstruction presented earlier is therefore obtained as the special case $N_m = 1$, for which the Galerkin system reduces exactly to the 2×2 interface system derived in Section 3.1. In practice, coefficient variability, numerical quadrature, interpolation errors, finite-element discretization, and geometric approximations introduce weak modal coupling. The coupled system (38) should therefore be regarded as the fundamental formulation, while the decoupled modal systems provide an efficient approximation whenever the basis remains sufficiently orthogonal.

3.2.5 Conditioning and Error Analysis

The reconstruction error depends primarily on two factors:

1. the magnitude of the errors present in the interface traces;
2. the conditioning of the reconstruction matrix $\mathbf{A}$.

Let α^* denote the exact modal amplitudes and α the reconstructed amplitudes. Introducing an interface perturbation vector ε , the Galerkin system becomes

$$\mathbf{A}\alpha = \mathbf{b} + \varepsilon.$$

The amplitude error therefore satisfies

$$\Delta\alpha = \alpha - \alpha^* = \mathbf{A}^{-1}\varepsilon,$$

which implies

$$\|\Delta\alpha\| \lesssim \kappa(\mathbf{A}) \|\varepsilon\|.$$

Consequently, poor conditioning amplifies errors in the interface traces and may lead to inaccurate reconstruction of the homogeneous correction. Unlike the overlap formulation, however, the reconstruction uses physically meaningful quantities directly at the interface. No observation-point selection or overlap-width parameter is required. The principal source of ill-conditioning is therefore not geometric redundancy but rather insufficient modal orthogonality or the inclusion of a large number of weakly excited modes. A useful a posteriori indicator is provided by the residuals

$$r_u = \|R_u\|_2, \quad r_q = \|R_q\|_2.$$

Small residuals indicate that both continuity conditions are satisfied accurately and that the recovered homogeneous correction successfully removes the interface inconsistency.

3.2.6 Corrected Solutions

Once Eq. (38) has been solved, the homogeneous corrections are reconstructed using Eq. (29). The corrected solutions are

$$\tilde{u}_W = u_W - u_h^{(W)}, \quad (39)$$

$$\tilde{u}_E = u_E - u_h^{(E)}. \quad (40)$$

The resulting fields satisfy continuity of both the solution and the normal flux across the artificial interface and therefore provide an approximation to the solution that would have been obtained on the original undecomposed domain.

3.2.7 2-Domains Test Case

The non-overlapping flux reconstruction methodology was first validated using a two-domain decomposition. The computational domain was partitioned into a western subdomain Ω_W and an eastern subdomain Ω_E , separated by a vertical artificial interface located at $x = x_I$. Artificial perturbations were imposed on the interface values of the independent subdomain solves, generating a homogeneous error that propagated into both subdomains. The reconstruction procedure was then applied using a truncated modal basis composed of four interface modes per subdomain. Unlike the overlap formulation, where information is collected throughout an overlap region, the present flux reconstruction uses only observations located directly on the interface. Consequently, only a small number of interface sampling points is required to determine the unknown modal amplitudes. In this experiment, the reconstruction employed two retained modes on each side of the interface, yielding four modal amplitudes per subdomain and therefore

$$N_{\text{unknowns}} = 8.$$

The interface continuity conditions were projected onto the same modal basis using a Galerkin formulation. Consequently, four modal equations were obtained from the solution residual and four modal equations from the flux residual, giving

$$N_{\text{equations}} = 8.$$

The resulting reconstruction matrix was therefore square,

$$\mathbf{A} \in \mathbb{R}^{8 \times 8}.$$

Unlike a pointwise least-squares formulation, the number of interface sampling points does not determine the size of the reconstruction system. The interface points are used only for numerical quadrature of the Galerkin integrals. Increasing the number of sampling points therefore improves the accuracy of the integral evaluation, but does not

change the number of unknown amplitudes or the size of the reconstruction matrix. The resulting reconstruction matrix remained well conditioned, with

$$\kappa(\mathbf{A}) = 2.78 \times 10^1. \quad (41)$$

The quality of the recovered homogeneous corrections was quantified by comparing the analytical reconstruction against the exact homogeneous error obtained from the reference solution. The resulting reconstruction errors were

$$\|u_h^{(W)} - \hat{u}_h^{(W)}\|_2 = 3.09 \times 10^{-1}, \quad (42)$$

$$\|u_h^{(E)} - \hat{u}_h^{(E)}\|_2 = 1.22 \times 10^{-1}. \quad (43)$$

Despite the extremely small number of interface observations, the reconstructed homogeneous fields accurately reproduced the dominant error modes generated by the artificial interface conditions. The global solution error before reconstruction was

$$\|e_{\text{before}}\|_2 = 1.83 \times 10^2, \quad (44)$$

while after reconstruction the error was reduced to

$$\|e_{\text{after}}\|_2 = 3.15 \times 10^{-1}. \quad (45)$$

This corresponds to an improvement factor of

$$\frac{\|e_{\text{before}}\|_2}{\|e_{\text{after}}\|_2} = 5.81 \times 10^2. \quad (46)$$

Figure 17 presents the error field before and after reconstruction. Prior to correction, the error is strongly concentrated near the artificial interface and reaches amplitudes approaching 10. After reconstruction, the interface error is almost completely removed and only a small residual remains, approximately three orders of magnitude smaller than the original perturbation. A particularly important observation is that the present non-overlapping formulation requires

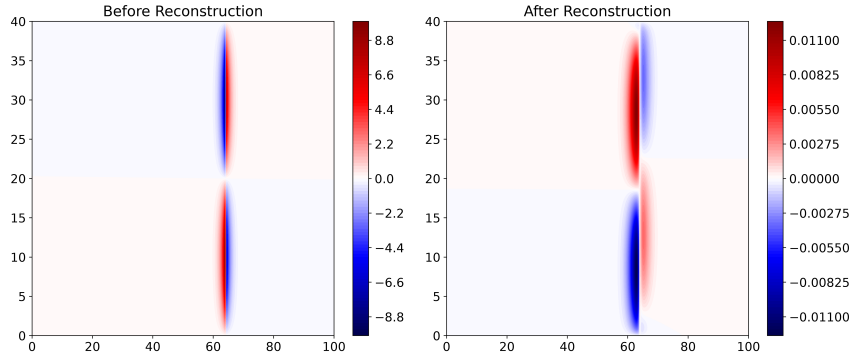


Figure 17: Error field before and after two-domain flux reconstruction. The reconstruction removes almost the entire interface-induced error using only four interface observation points.

significantly fewer observations than the corresponding overlap reconstruction. In the overlap approach, the homogeneous amplitudes must be recovered from measurements distributed throughout a finite two-dimensional overlap region. As a consequence, a substantially larger number of observation points is typically required to obtain a stable least-squares system. Furthermore, observations located far from the interface may contain little information about the homogeneous component and can introduce additional numerical noise. In contrast, the flux formulation uses only interface quantities, namely the solution and normal flux jumps. These quantities contain the information most directly associated with the homogeneous correction. The reconstruction therefore extracts the modal amplitudes from a much smaller number of samples while maintaining excellent conditioning and accuracy. The results demonstrate that the interface-based reconstruction is considerably more information-efficient than the overlap approach. With only four interface sampling locations, corresponding to eight continuity constraints in total, the method reduces the global error by more than two and a half orders of magnitude while maintaining a condition number below 30. This indicates that the interface conditions alone provide sufficient information to accurately recover the dominant homogeneous error modes in the two-domain configuration.

3.3 Conclusion and Remarks

This chapter introduced a homogeneous-error reconstruction framework for non-overlapping domain decompositions. In contrast to the overlap formulation presented in the previous chapter, the reconstruction no longer relies on observations collected inside a common overlap region. Instead, all information is extracted directly from the artificial interfaces separating neighbouring subdomains through continuity of the solution and continuity of the normal flux. The fundamental observation is that the interface mismatch generated by independent subdomain solves belongs entirely to the homogeneous solution space. By representing the homogeneous correction using the analytical modal basis developed previously, the unknown amplitudes can be recovered from interface residuals alone. This leads naturally to a Galerkin reconstruction problem in which the interface jumps are projected onto a set of interface modes. For general multidomain configurations the resulting formulation is expressed as a global interface system. In the important case of rectangular domains with approximately separable coefficients, modal orthogonality causes the reconstruction matrix to become nearly block diagonal. The reconstruction then reduces to a sequence of small independent systems, providing a computationally efficient approximation of the full interface problem. The numerical results demonstrate that the method is capable of recovering the dominant homogeneous error modes with high accuracy. In the two-domain validation case, the reconstruction reduced the global error from 1.83×10^2 to 3.15×10^{-1} , corresponding to an improvement factor of approximately 5.81×10^2 . At the same time, the homogeneous corrections recovered on the western and eastern subdomains remained in close agreement with the exact homogeneous components extracted from the reference solution. An important advantage of the flux formulation is the amount of information required for reconstruction. In the overlap approach, the homogeneous amplitudes are determined from observations distributed throughout a finite two-dimensional overlap region. The number of observation points therefore increases with overlap size, and the resulting least-squares systems may contain significant redundancy. Furthermore, observations located far from the interface often contain limited information about the homogeneous component and may contribute to ill-conditioning. The present non-overlapping formulation is fundamentally different. The information used for reconstruction is concentrated directly on the interface, which is precisely where the homogeneous error originates. Consequently, a relatively small number of interface sampling points is sufficient to recover the dominant correction amplitudes. The two-domain test case required only four interface points, producing eight continuity constraints in total. Despite this extremely compact system, the reconstruction achieved excellent accuracy while maintaining a modest condition number of approximately

$$\kappa(\mathbf{A}) = 2.78 \times 10^1.$$

This behaviour contrasts with several overlap reconstructions presented previously, where large numbers of observations were often required to achieve comparable accuracy and where the conditioning could deteriorate significantly as the observation set became nearly linearly dependent. Another important observation is that increasing the number of interface points transforms the reconstruction into an increasingly overdetermined least-squares problem. This provides a natural mechanism for improving robustness in the presence of discretization errors, interpolation errors, or numerical noise. The interface sampling density therefore plays a role analogous to the number of observation points in the overlap methodology. A constraint-convergence study confirms that the reconstruction error decreases rapidly as additional interface constraints are introduced, eventually approaching a plateau determined by the modal truncation error and the finite-element discretization. Overall, the results demonstrate that homogeneous reconstruction remains effective even when no overlap region is available. By exploiting solution and flux continuity directly at the interface, the proposed methodology eliminates the need for overlap regions, observation-domain selection, and inverse reconstruction over extended portions of the computational domain. The resulting framework is local, inexpensive, naturally suited to disjoint domain decompositions, and scales directly to multidomain configurations involving multiple interfaces. The non-overlapping flux reconstruction and the overlap reconstruction should therefore be viewed as complementary formulations based on the same underlying homogeneous-error representation. The overlap formulation leverages redundant information distributed throughout a common observation region, whereas the flux formulation exploits the physically meaningful interface quantities where the homogeneous error is generated. For problems in which overlap regions are unavailable or undesirable, the flux-based approach provides a robust and efficient alternative while requiring substantially fewer observations and maintaining favourable conditioning properties.

4 Conclusion and Remarks

The objective of this work was to develop and investigate reconstruction techniques capable of recovering the solution of elliptic problems on decomposed computational domains without requiring additional global solves. Motivated by applications within the Variational Boussinesq Model (VBM), the central idea was to exploit the decomposition of the solution into particular and homogeneous components and to interpret the error introduced by artificial subdomain boundaries as a homogeneous correction. By reconstructing this correction and removing it from the independently computed subdomain solutions, it becomes possible to recover an approximation of the solution that would have been obtained on the original undecomposed domain. Two reconstruction frameworks were investigated. The first considered overlapping domain decompositions, where neighbouring subdomains share a common observation region.

The second considered disjoint domain decompositions, where no overlap is present and information is available only through the artificial interfaces. For overlapping decompositions, the reconstruction procedure was formulated as an amplitude-recovery problem in which the homogeneous corrections are identified from differences between neighbouring subdomain solutions within the overlap region. In one dimension, the homogeneous correction can be represented using only two exponentially decaying basis functions. Numerical experiments showed that the dominant interface error can be recovered accurately even when the overlap is reduced to the minimum configuration corresponding to only a single physical grid-point layer. Global error reductions of approximately two orders of magnitude were obtained while employing only two overlap observations. These results demonstrate that the essential information required to reconstruct the homogeneous correction is already contained within extremely small overlap regions. The extension to two spatial dimensions revealed a fundamentally richer reconstruction problem. Rather than a small number of exponential corrections, the homogeneous solution space contains multiple admissible modes associated with the eigenstructure of the governing elliptic operator. The reconstruction therefore becomes a modal-amplitude recovery problem. The numerical experiments demonstrated that relatively few modes are sufficient to capture the dominant interface-induced error. However, the studies also revealed that reconstruction accuracy is governed not only by modal completeness but also by the conditioning of the underlying inverse problem. Increasing the number of retained modes alone does not necessarily improve accuracy if the associated amplitudes cannot be recovered robustly from the available observations. A major outcome of the overlap investigations is the identification of the overlap region itself as a critical factor controlling reconstruction performance. When the overlap width becomes small, observations are collected at nearly identical distances from the artificial interfaces, producing highly correlated information and therefore poorly conditioned reconstruction matrices. Increasing the overlap width substantially improves the independence of the observations and consequently improves the recovery of the modal amplitudes. The numerical studies consistently showed that reconstruction accuracy depends on a balance between modal representation, conditioning, and geometric information content. The four-domain overlap experiments demonstrated that the methodology extends naturally beyond simple two-domain configurations. Even in the presence of multiple interacting interfaces and several simultaneous families of homogeneous modes, the dominant interface error could still be recovered accurately. Global improvements exceeding two orders of magnitude were obtained while using only a small number of modal amplitudes and observations. These results indicate that the reconstruction process scales naturally to multidomain configurations of the type encountered in practical VBM simulations. The second part of this work addressed the significantly more challenging problem of non-overlapping domain decompositions. In this setting no common observation region exists and the reconstruction can no longer be formulated in terms of overlap differences. Instead, the homogeneous correction was recovered directly from continuity of the solution and continuity of the normal flux at the artificial interface. For the one-dimensional problem, this approach leads to a remarkably simple reconstruction system. The unknown correction amplitudes are obtained from a 2×2 interface system that depends only on the solution and flux jumps. A theoretical analysis showed that the conditioning of this system remains favourable even for very large computational domains. Numerical experiments confirmed this prediction and demonstrated reconstruction factors exceeding three orders of magnitude while maintaining condition numbers of only $\mathcal{O}(10^1)$. The two-dimensional flux formulation generalized the reconstruction procedure through a Galerkin projection of the interface residuals onto the homogeneous modal basis. Unlike the overlap approach, where observations are distributed throughout a two-dimensional region, the flux formulation uses only information located directly on the interface itself. An important consequence is that the reconstruction requires significantly fewer observations. In the two-domain test case, only four interface sampling points were sufficient to recover the dominant modal amplitudes. Despite using far less information than the corresponding overlap formulation, the method reduced the global error by more than two orders of magnitude while maintaining excellent conditioning. This difference highlights a fundamental distinction between the two approaches. The overlap reconstruction operates as an inverse problem in which the homogeneous correction is inferred from field observations distributed throughout an overlap region. The flux reconstruction, by contrast, exploits the interface quantities that are directly responsible for generating the homogeneous error. Consequently, it often requires substantially less information and exhibits notably better conditioning. Taken together, the results of this work demonstrate that homogeneous-error reconstruction provides a practical and effective framework for coupling independently solved subdomains. Both overlap-based and flux-based formulations are capable of recovering the dominant interface-induced errors without performing additional global elliptic solves. The overlap formulation is particularly attractive when overlap regions are already present within the computational framework, whereas the flux formulation provides a natural alternative for disjoint decompositions and exhibits superior information efficiency.

Future Work

Several directions for future research arise naturally from the developments presented in this work. First, the present investigation was restricted primarily to rectangular domains and problems admitting analytical modal descriptions through separation of variables. An important extension would be the development of numerical eigenfunction bases for irregular geometries, variable coefficients, and realistic coastal bathymetries. Such an extension would significantly increase the applicability of the methodology within practical engineering simulations. Second, the multidomain

studies considered only relatively small numbers of subdomains. Future work should investigate larger multidomain decompositions containing many interfaces and cross-points. In particular, it would be valuable to study how the conditioning, modal interactions, and reconstruction accuracy scale with increasing numbers of subdomains. Third, the overlap and non-overlap formulations suggest different approaches to observation selection. For overlap reconstruction, the location and distribution of observations strongly influence conditioning. Future research could investigate adaptive observation strategies based on error estimates, reduced-basis techniques, optimal experimental design, or machine-learning-assisted sampling. Such approaches could automatically determine the minimum amount of information required for accurate reconstruction. Fourth, the flux formulation naturally raises the question of constraint selection. The present study showed that only a small number of interface points is sufficient for accurate reconstruction. A systematic analysis of reconstruction error as a function of interface sampling density, modal resolution, and interface conditioning would provide valuable design guidelines for practical implementations. Fifth, the present work focused exclusively on linear elliptic problems. Extending the reconstruction methodology to time-dependent, nonlinear, or multiphysics systems represents an important future direction. In particular, the governing equations arising within modern wave models contain nonlinear effects that may alter the behaviour of the homogeneous correction and its associated modal structure. Another promising direction concerns the integration of the reconstruction procedure directly into iterative domain decomposition methods. Rather than applying the correction as a post-processing step, the reconstructed homogeneous field could be used to improve interface transmission conditions or accelerate convergence of Schwarz-type methods. Finally, the methodology was developed with the Variational Boussinesq Model in mind. A complete integration of the reconstruction framework within the operational VBM-XMF environment would provide the ultimate validation of the approach. Such studies would quantify potential reductions in computational cost, memory usage, communication overhead, and scalability for realistic large-scale maritime simulations. Overall, the results obtained in this work demonstrate that homogeneous reconstruction offers a promising alternative to traditional interface-coupling strategies. By exploiting the structure of the governing differential operator, it is possible to recover highly accurate approximations of the global solution using only local subdomain information. The consistent improvements observed for both overlapping and disjoint decompositions suggest that homogeneous reconstruction provides a strong foundation for future developments in efficient multidomain elliptic solvers and real-time wave simulation technology.

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A MARIN VBM-XMF Computational Grid

The computational-grid notation and indexing convention adopted in the MARIN VBM-XMF implementation follow the Variational Boussinesq Model (VBM) described by Klopman (Ref. [6]). The description presented in this appendix is restricted to the computational grid relevant to the present work. Figure 18 shows the structured Cartesian grid used

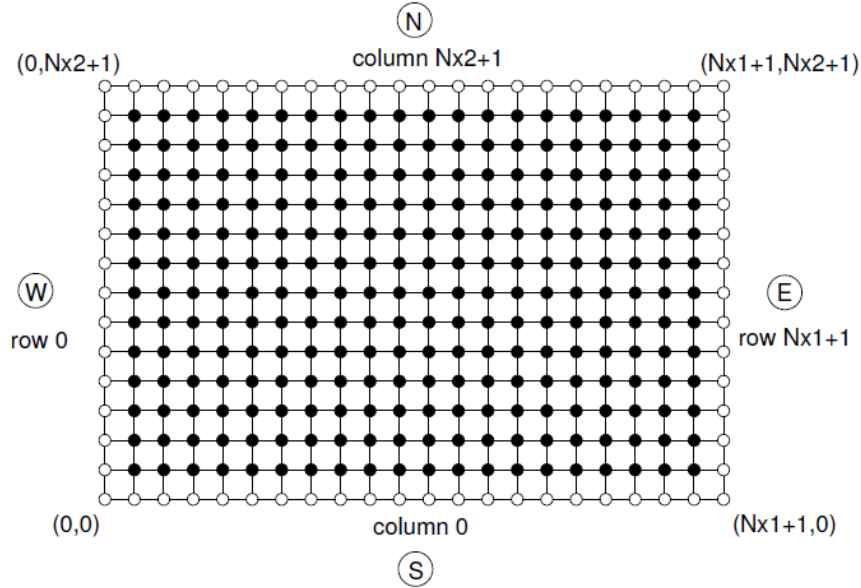


Figure 18: Computational grid employed by the Variational Boussinesq Model (VBM), adapted from Ref. [6]. Black nodes denote the physical computational domain, whereas white nodes denote ghost nodes used for boundary treatment and the efficient solution of elliptic equations.

by the VBM. The interior black nodes represent the physical fluid domain, while an additional layer of ghost nodes surrounds the domain. These ghost nodes are located outside the physical domain and facilitate the implementation of boundary conditions and numerical operators. The physical computational domain consists of the interior indices $i_1 = 1, \dots, N_{x_1}$ and $i_2 = 1, \dots, N_{x_2}$. The outer ghost-node layers correspond to row 0, row $N_{x_1} + 1$, column 0, and column $N_{x_2} + 1$. Consequently, the complete computational stencil contains $(N_{x_1} + 2) \times (N_{x_2} + 2)$ nodes. The coordinates of the grid nodes are given by

$$x_1 = (i_1 - 1)\Delta x_1, \quad i_1 = 0, \dots, N_{x_1} + 1, \quad (47)$$

and

$$x_2 = (i_2 - 1)\Delta x_2, \quad i_2 = 0, \dots, N_{x_2} + 1. \quad (48)$$

Here, Δx_1 and Δx_2 denote the constant grid spacing in the x_1 - and x_2 -directions. Defining the domain lengths

$$L_1 = (N_{x_1} - 1)\Delta x_1, \quad L_2 = (N_{x_2} - 1)\Delta x_2, \quad (49)$$

the physical computational domain is

$$(x_1, x_2) \in [0, L_1] \times [0, L_2]. \quad (50)$$

Accordingly, the physical boundaries correspond to $x_1 = 0$ when $i_1 = 1$, $x_1 = L_1$ when $i_1 = N_{x_1}$, $x_2 = 0$ when $i_2 = 1$, and $x_2 = L_2$ when $i_2 = N_{x_2}$. The corner ghost nodes shown in Figure 18 are indexed by $(0, 0)$, $(N_{x_1} + 1, 0)$, $(0, N_{x_2} + 1)$, and $(N_{x_1} + 1, N_{x_2} + 1)$. For orientation purposes, the numerical implementation adopts the North–South–East–West convention indicated in Figure 18. In terms of grid indices, increasing i_1 corresponds to the east direction and increasing i_2 corresponds to the north direction. Thus, the east, west, north, and south neighbours of node (i_1, i_2) are given by $(i_1 + 1, i_2)$, $(i_1 - 1, i_2)$, $(i_1, i_2 + 1)$, and $(i_1, i_2 - 1)$, respectively.

B Two-Dimensional Homogeneous Modes and Modal Basis Functions

This appendix derives the homogeneous eigenfunctions and modal basis functions employed throughout the two-dimensional reconstruction framework. The resulting basis functions are used both for overlapping-domain reconstruction (Section 2) and for the disjoint-domain reconstruction methodology introduced in Section 3.

The objective of the appendix is to identify a family of homogeneous solutions that can represent the interface error generated by artificial domain decomposition. These modes provide a natural basis for reconstructing homogeneous corrections associated with artificial interfaces.

B.1 Homogeneous Problem

Consider the elliptic operator introduced in Eq. (1). The homogeneous problem is

$$ku - \nabla \cdot (h \nabla u) = 0. \quad (51)$$

Within a sufficiently small overlap region, the coefficient h is assumed to vary slowly and may therefore be approximated by a constant representative value. Equation (51) then becomes

$$-h \Delta u + ku = 0. \quad (52)$$

The purpose of this appendix is to characterize the solution space of Eq. (52).

B.2 Separation of Variables

Consider a rectangular domain

$$\Omega = [0, L_x] \times [0, L_y].$$

Seeking separable solutions of the form

$$u(x, y) = X(x)Y(y),$$

and substituting into Eq. (52) gives

$$-h(X''Y + XY'') + kXY = 0.$$

Dividing by XY yields

$$-h \left(\frac{X''}{X} + \frac{Y''}{Y} \right) + k = 0.$$

Rearranging gives

$$-\frac{X''}{X} - \frac{Y''}{Y} + \frac{k}{h} = 0.$$

Introducing a separation constant leads to

$$-\frac{Y''}{Y} = \beta_n^2,$$

and

$$\frac{X''}{X} = \frac{k}{h} + \beta_n^2.$$

B.3 Transverse Eigenfunctions

The transverse equation becomes

$$Y'' + \beta_n^2 Y = 0.$$

Imposing homogeneous boundary conditions on

$$y = 0, \quad y = L_y,$$

gives

$$Y_n(y) = \sin\left(\frac{n\pi y}{L_y}\right),$$

with eigenvalues

$$\beta_n = \frac{n\pi}{L_y}, \quad n = 1, 2, \dots$$

The corresponding eigenfunctions form an orthogonal basis on $[0, L_y]$.

B.4 Longitudinal Behaviour

The longitudinal problem becomes

$$X_n'' - \alpha_n^2 X_n = 0,$$

where

$$\alpha_n = \sqrt{\frac{k}{h} + \left(\frac{n\pi}{L_y}\right)^2}. \quad (53)$$

The general solution is

$$X_n(x) = C_1 e^{\alpha_n x} + C_2 e^{-\alpha_n x},$$

or equivalently

$$X_n(x) = A_n \sinh(\alpha_n x) + B_n \cosh(\alpha_n x).$$

Consequently, the homogeneous modes become

$$u_n(x, y) = \sinh(\alpha_n(L_x - x)) \sin\left(\frac{n\pi y}{L_y}\right). \quad (54)$$

The first four modal structures are shown in Figure 5.

These figures illustrate a key difference from the one-dimensional problem. In one dimension only a small number of decay modes are available. In two dimensions, however, multiple transverse eigenfunctions contribute simultaneously, resulting in a genuinely multimodal reconstruction space.

B.5 Modal Representation of the Homogeneous Correction

The homogeneous correction may therefore be expanded as

$$u_h(x, y) = \sum_{n=1}^{\infty} a_n \phi_n(x, y),$$

where a_n denotes the amplitude associated with the n -th mode. In practical applications only a finite number of modes contribute significantly. The reconstruction therefore employs the truncated expansion

$$u_h(x, y) \approx \sum_{n=1}^{N_m} a_n \phi_n(x, y), \quad (55)$$

where N_m denotes the number of retained modes.

B.6 Directional Basis Functions

The previous derivation considered only one interface orientation. In practical computations, artificial interfaces may occur on the western, eastern, northern, or southern boundaries of a subdomain. Following the directional convention adopted throughout the report, four modal families are introduced.

Consider a rectangular subdomain $\Omega = [a, b] \times [c, d]$. Define

$$L_x = b - a, \quad L_y = d - c,$$

and

$$\mu = \sqrt{\frac{k}{h}}.$$

B.6.1 Western Modes

For a western interface the modal basis is

$$\phi_n^{(W)} = \frac{\sinh(\alpha_n(b-x))}{\sinh(\alpha_n L_x)} \sin(\beta_n(y-c)), \quad (56)$$

where

$$\beta_n = \frac{n\pi}{L_y}, \quad \alpha_n = \sqrt{\mu^2 + \beta_n^2}.$$

These functions satisfy

$$\phi_n^{(W)} = 1$$

at the artificial western interface and decay into the interior.

B.6.2 Eastern Modes

For an eastern interface,

$$\phi_n^{(E)} = \frac{\sinh(\alpha_n(x-a))}{\sinh(\alpha_n L_x)} \sin(\beta_n(y-c)). \quad (57)$$

These modes are normalized at the eastern interface and decay away from it.

B.6.3 Southern Modes

For interfaces normal to the y -direction,

$$\phi_n^{(S)} = \frac{\sinh(\gamma_n(d-y))}{\sinh(\gamma_n L_y)} \sin\left(\frac{n\pi(x-a)}{L_x}\right), \quad (58)$$

with

$$\gamma_n = \sqrt{\mu^2 + \left(\frac{n\pi}{L_x}\right)^2}.$$

B.6.4 Northern Modes

Similarly,

$$\phi_n^{(N)} = \frac{\sinh(\gamma_n(y-c))}{\sinh(\gamma_n L_y)} \sin\left(\frac{n\pi(x-a)}{L_x}\right). \quad (59)$$

C Boundary-Source Representation and Modal Projection

The reconstruction procedures introduced throughout this work recover the modal amplitudes from observations collected within the overlap region. An alternative interpretation is obtained by considering the boundary error itself as the source of the homogeneous correction. Consider a subdomain possessing one or more artificial boundaries. Let the interface error imposed on a boundary be denoted by

$$e_\Gamma = u_\Gamma - u_\Gamma^{\text{exact}},$$

where u_Γ denotes the imposed boundary data and u_Γ^{exact} denotes the trace of the solution that would be obtained on the undecomposed domain. The homogeneous correction generated by the interface may then be interpreted as the solution of

$$ku_h - \nabla \cdot (h \nabla u_h) = 0$$

subject to the artificial boundary forcing

$$u_h = e_\Gamma \quad \text{on } \Gamma.$$

The interface error therefore acts as a boundary source for the homogeneous correction. In practical applications, the interface error is available only at a finite collection of grid locations,

$$\{(x_i, y_i)\}_{i=1}^{N_s},$$

located on the artificial boundaries

$$x = x_W, \quad x = x_E, \quad y = y_S, \quad y = y_N.$$

These values may be regarded as a discrete set of source points that generate the homogeneous correction within the neighbouring domains. For a western interface, the boundary error may be expanded in the transverse eigenfunctions,

$$e_W(y) = \sum_{n=1}^{\infty} a_n^{(W)} \sin\left(\frac{n\pi(y-c)}{L_y}\right).$$

Using the orthogonality of the sine basis gives

$$a_n^{(W)} = \frac{2}{L_y} \int_c^d e_W(y) \sin\left(\frac{n\pi(y-c)}{L_y}\right) dy.$$

Similarly,

$$a_n^{(E)} = \frac{2}{L_y} \int_c^d e_E(y) \sin\left(\frac{n\pi(y-c)}{L_y}\right) dy,$$

for eastern interfaces and

$$a_n^{(S)} = \frac{2}{L_x} \int_a^b e_S(x) \sin\left(\frac{n\pi(x-a)}{L_x}\right) dx,$$

$$a_n^{(N)} = \frac{2}{L_x} \int_a^b e_N(x) \sin\left(\frac{n\pi(x-a)}{L_x}\right) dx,$$

for southern and northern interfaces. In practice the boundary error is usually available only at discrete source locations. The projection integrals are therefore replaced by quadrature rules,

$$a_n^{(W)} \approx \frac{2}{N_s} \sum_{k=1}^{N_s} e_W(y_k) \sin\left(\frac{n\pi(y_k-c)}{L_y}\right),$$

with analogous expressions for the remaining boundaries. After computing the modal amplitudes, the homogeneous correction is reconstructed by

$$u_h = \sum_{n=1}^{N_m^W} a_n^{(W)} \phi_n^{(W)} + \sum_{n=1}^{N_m^E} a_n^{(E)} \phi_n^{(E)} + \sum_{n=1}^{N_m^N} a_n^{(N)} \phi_n^{(N)} + \sum_{n=1}^{N_m^S} a_n^{(S)} \phi_n^{(S)}.$$

The resulting approach differs from the overlap-observation formulation of Section 2. Rather than inferring the modal amplitudes from the field difference inside the overlap region, the amplitudes are obtained directly from the boundary error itself. In this sense, the artificial boundary values act as a collection of source points whose strengths are projected onto the homogeneous eigenfunctions. The formulation is particularly attractive for very narrow overlaps and forms a natural bridge toward the disjoint-domain reconstruction procedures introduced in Section 3, where interface information is available primarily through interface traces rather than overlapping solution values.